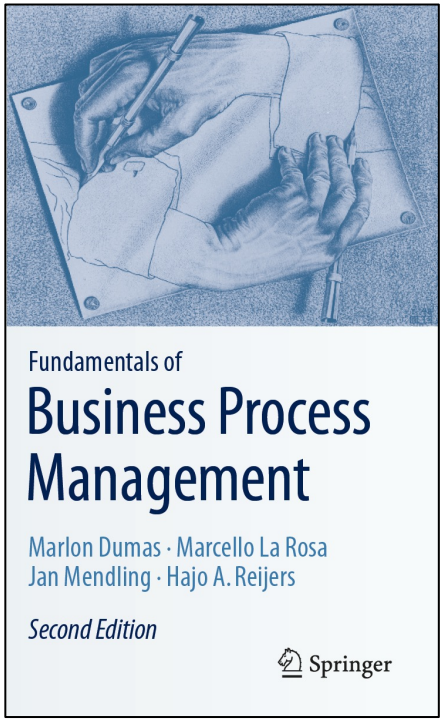
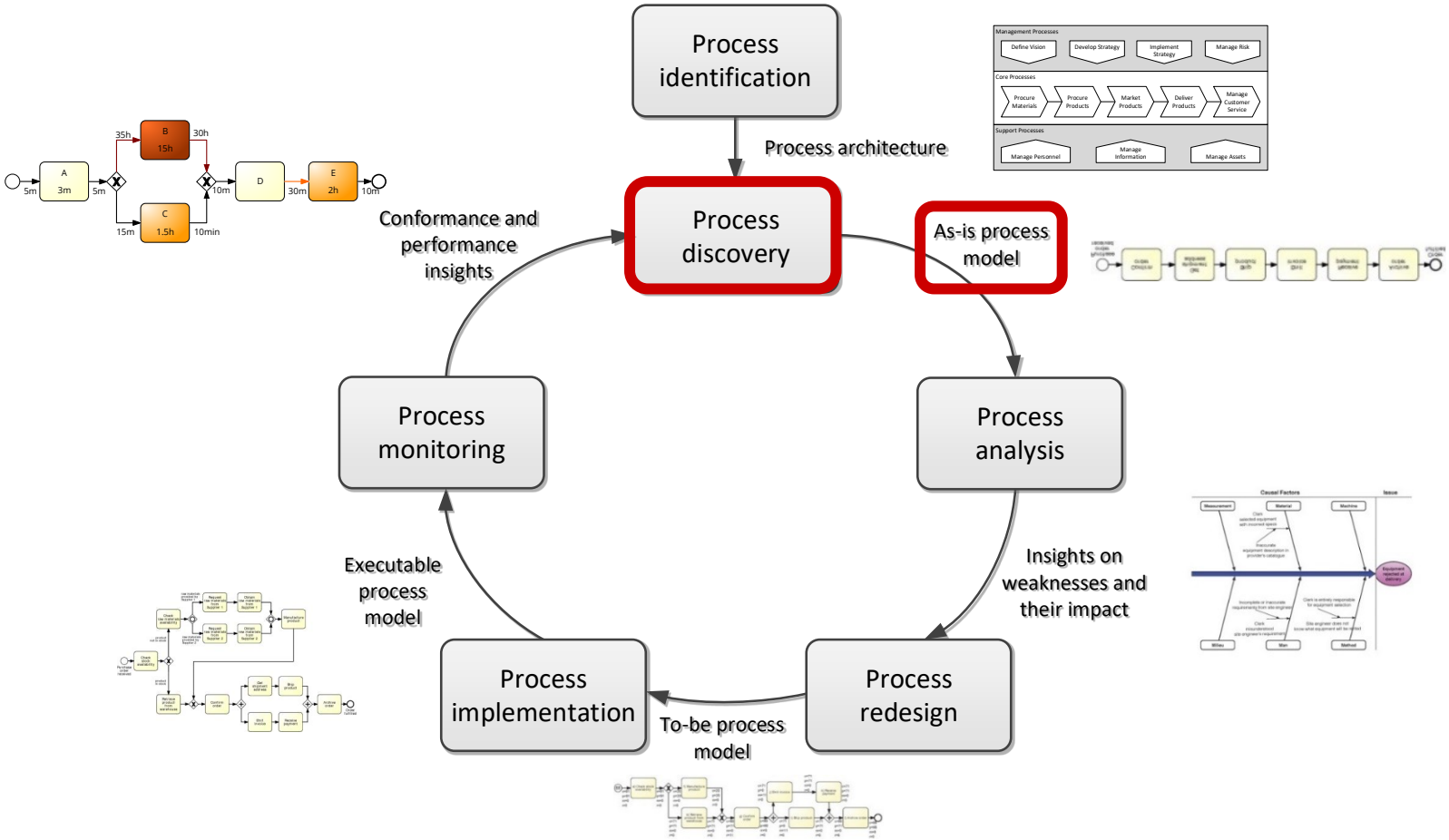
Chapter 3: Essential Process Modeling



Contents

1. First Steps with BPMN
2. Branching and Merging
3. Business Objects
4. Resources
5. Process Decomposition
6. Process Model Reuse
7. Recap

Process Modeling in the BPM Lifecycle



Business Process Model and Notation (BPMN)

- OMG standard (BPMN 2.1)
- Supported by numerous tools: bpmn.org lists over 70 tools
- Both for conceptual and executable models

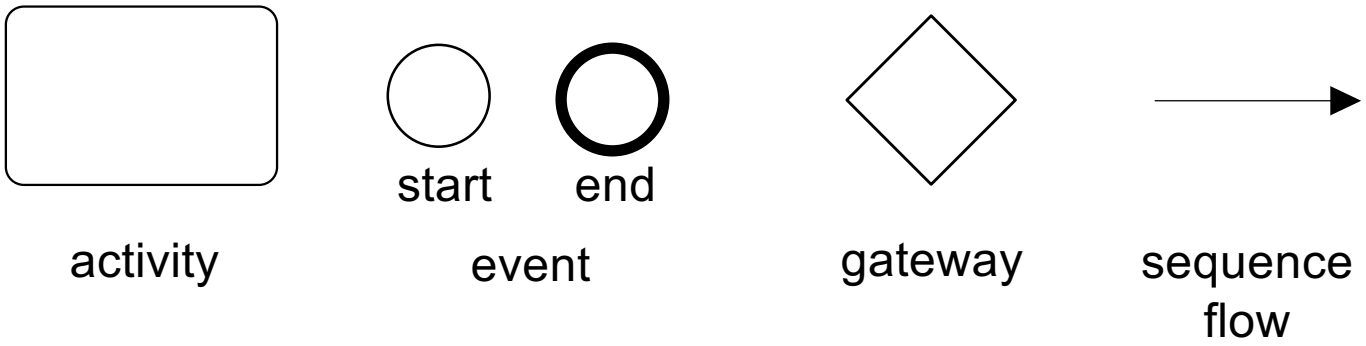


BPMN from 10,000 miles…

Based on popular graphical flowcharts:

- Core set of notation elements
- Each core element has various subtypes

A BPMN process model is a graph consisting of four types of **core elements**:



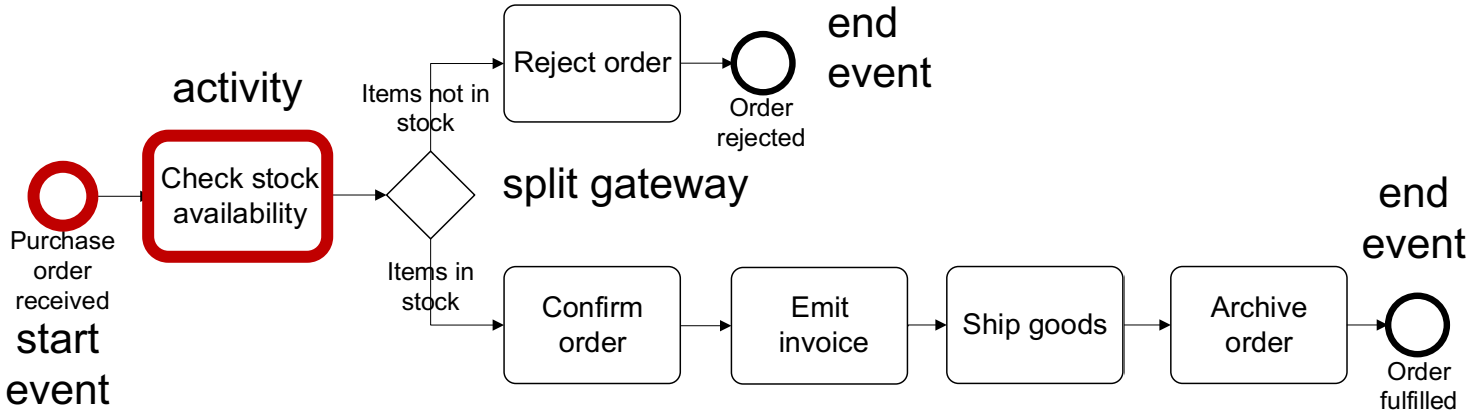
Let’s start modeling

Order-to-cash

A typical order-to-cash process is triggered by the receipt of a purchase order from a customer. The purchase order has to be checked against the stock regarding the availability of the item(s) requested. Depending on stock availability the purchase order may be confirmed or rejected.

If the purchase order is confirmed, an invoice is emitted and the goods requested are shipped. The process completes by archiving the order or if the order is rejected.

Solution in BPMN: Order-to-cash



Naming conventions

- Event: noun + past-participle verb (e.g. insurance claim lodged)
- Activity: imperative verb + noun (e.g. assess credit risk)

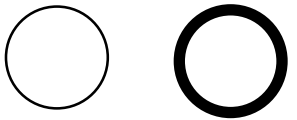
BPMN core elements



activity

Activities capture work performed in a process

- Different types of activities



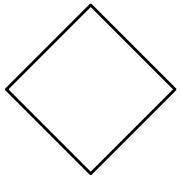
start event

end event

Events represent the process' triggers (start event) and outcomes (end event).

- Different types of events

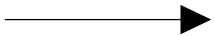
BPMN core elements



gateway

Gateways capture forking and joining paths in the control flow.

- Different types of gateways



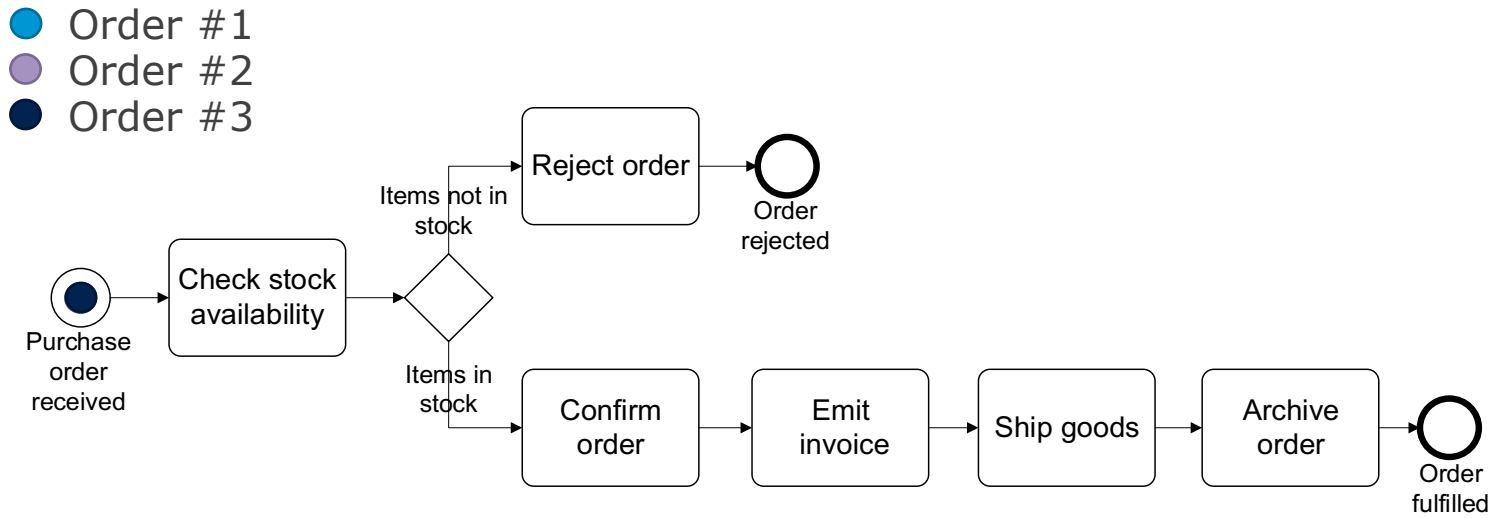
sequence flow

Sequence flows represent the order in which activities and events will be performed.

They can be assigned a condition to distinguish between alternative branches.

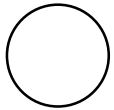
- Different types of flows

Process model vs process instances: The tokens game



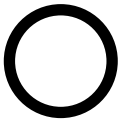
A little bit more on events…

A *start event* triggers a new process instance by generating a token that traverses the sequence flow (“tokens source”)



start event

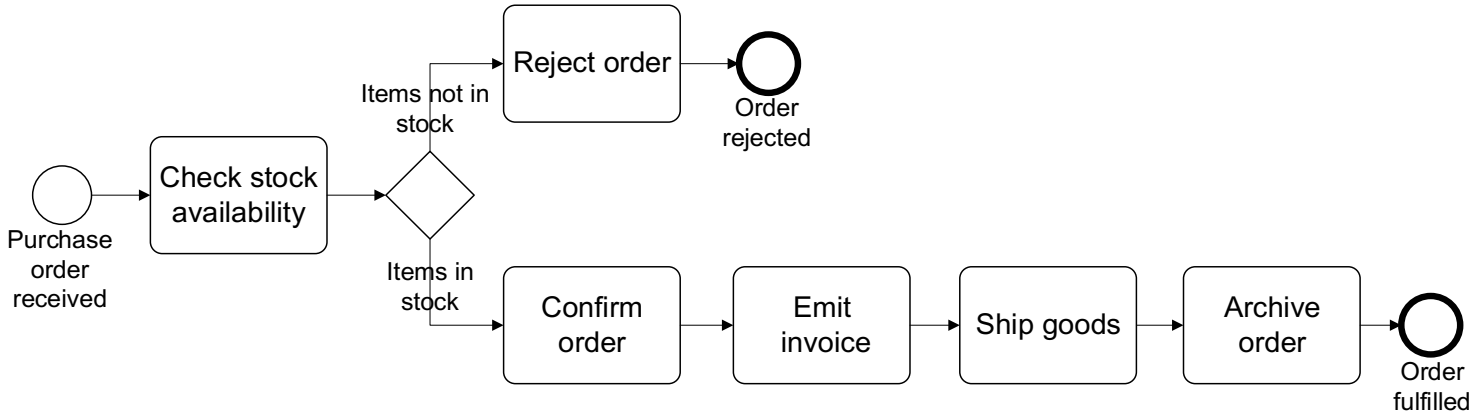
An *end event* signals that a process instance has completed with a given outcome by consuming a token (“tokens sink”)



end event

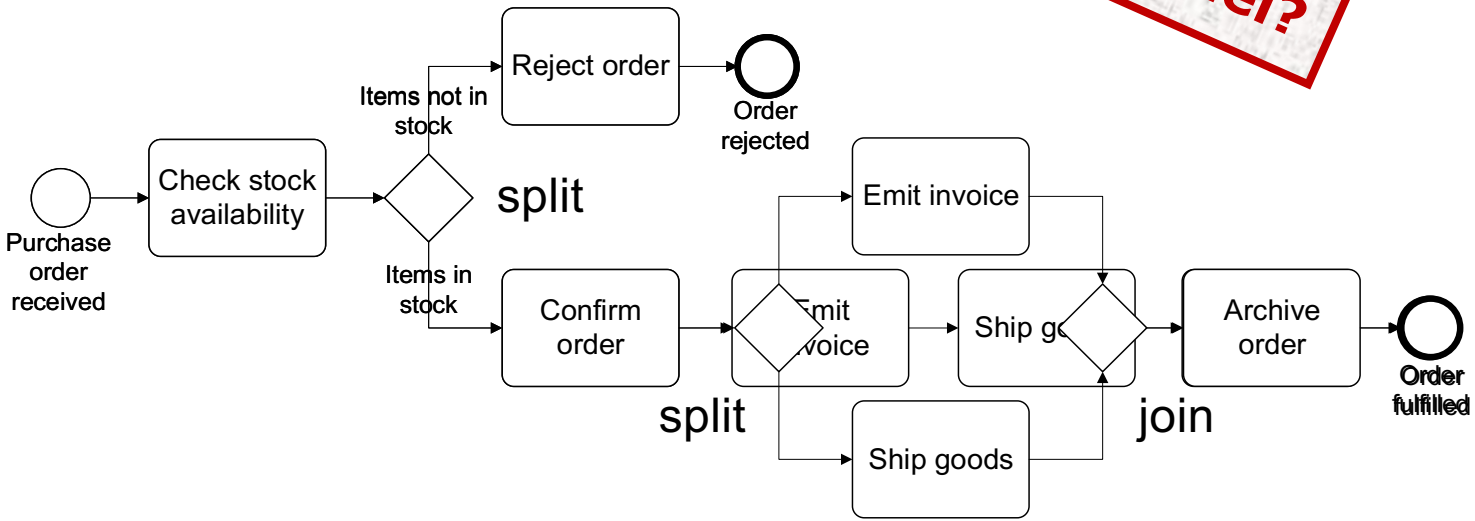
Let's reconsider our order-to-cash example

[…] If the purchase order is confirmed, **an invoice is emitted and the goods requested are shipped**. The process completes by archiving the order. […]



Solution: Order-to-cash

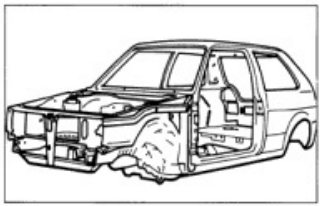
Anything wrong with this model?



Mapping, Abstraction, and Purpose of a Model



Models are **abstractions from real world phenomena**, developed for the purpose of **reducing overall complexity**.



Models **aggregate information** and **document only relevant aspects** of the real world

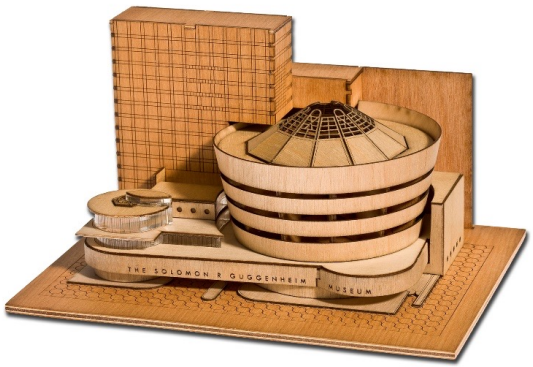


Models are being developed:

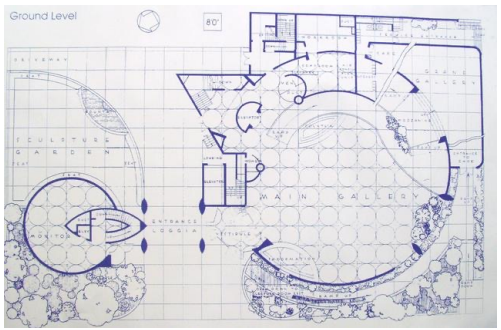
1. in a specific **modelling subject**
2. for a specific **target audience**
3. with a specific **modelling purpose** in mind

What's the relevant model?





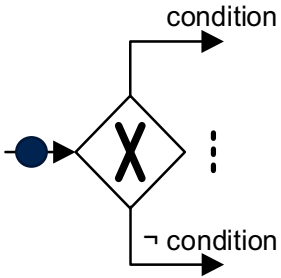
?



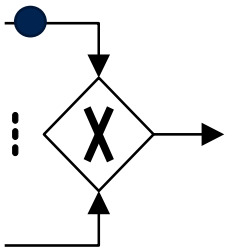
A little more on gateways: XOR Gateway



An *XOR Gateway* captures decision points (XOR-split) and points where alternative flows are merged (XOR-join)



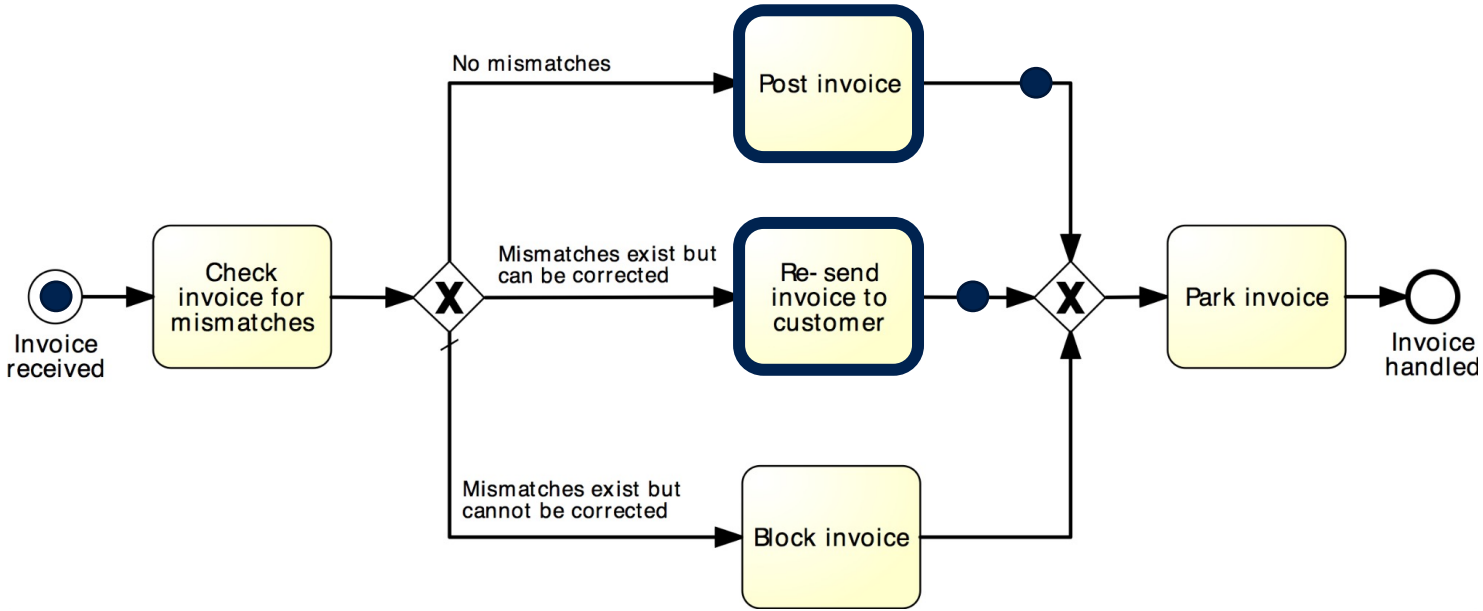
XOR-split ➔ takes **one** outgoing branch



XOR-join ➔ proceeds when **one** incoming branch has completed

Example: XOR Gateway

Invoice checking process

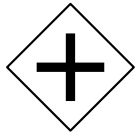


Exercise 3.1

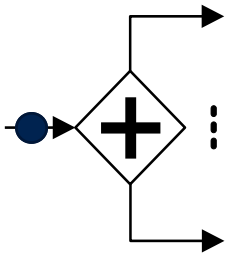
Model the following fragment of a business process for assessing loan applications (loan origination process).

Once a loan application has been approved by the loan provider, an acceptance pack is prepared and sent to the customer. The acceptance pack includes a repayment schedule which the customer needs to agree upon by sending the signed documents back to the loan provider. The latter then verifies the repayment agreement: if the applicant disagreed with the repayment schedule, the loan provider cancels the application; if the applicant agreed, the loan provider approves the application. In either case, the process completes with the loan provider notifying the applicant of the application status.

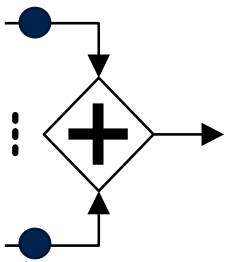
A little more on gateways: AND Gateway



An *AND Gateway* provides a mechanism to create and synchronize “parallel” flows.



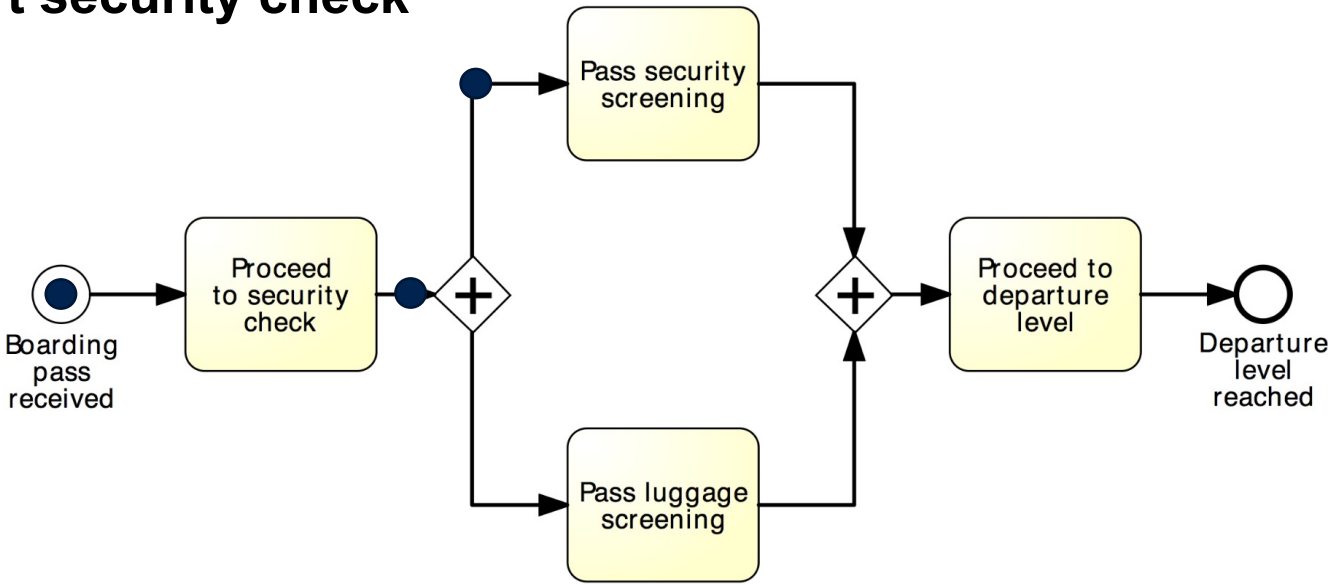
AND-split ➔ takes **all** outgoing branches



AND-join ➔ proceeds when **all** incoming branches have completed

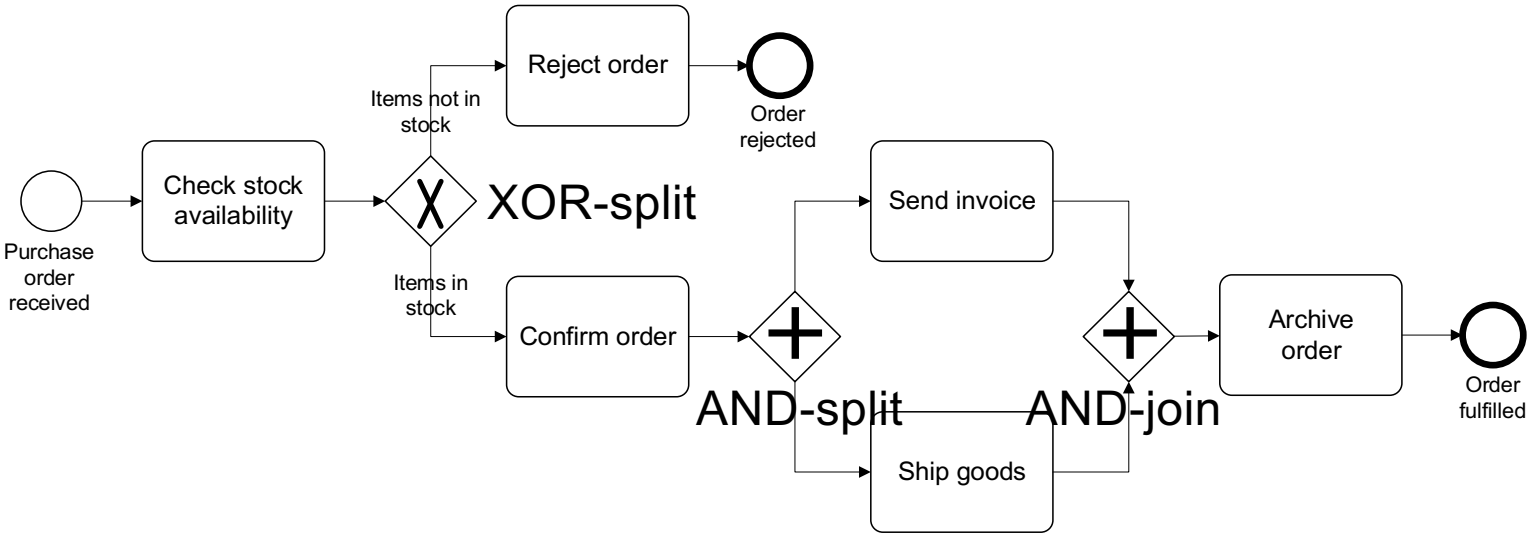
Example: AND Gateway

Airport security check



Revised solution

Order-to-cash

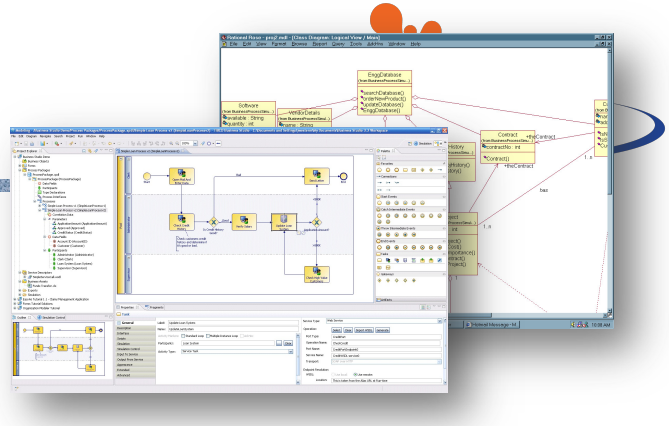


Exercise 3.2

Model the following fragment of a business process for assessing loan applications.

A loan application is approved if it passes two checks: i) the applicant's loan risk assessment, done automatically by a system, and ii) the appraisal of the property for which the loan has been asked, carried out by a property appraiser. The risk assessment requires a credit history check on the applicant, which is performed by a financial officer. Once both the loan risk assessment and the property appraisal have been performed, a loan officer can assess the applicant's eligibility. If the applicant is not eligible, the application is rejected, otherwise the acceptance pack is prepared and sent to the applicant.

Business Process Modeling Tools



Pen & Paper

Haptic

Standalone

- E.g. Visio, Camunda Modeler

Repository-based

- E.g. ARIS, Signavio, Apromore

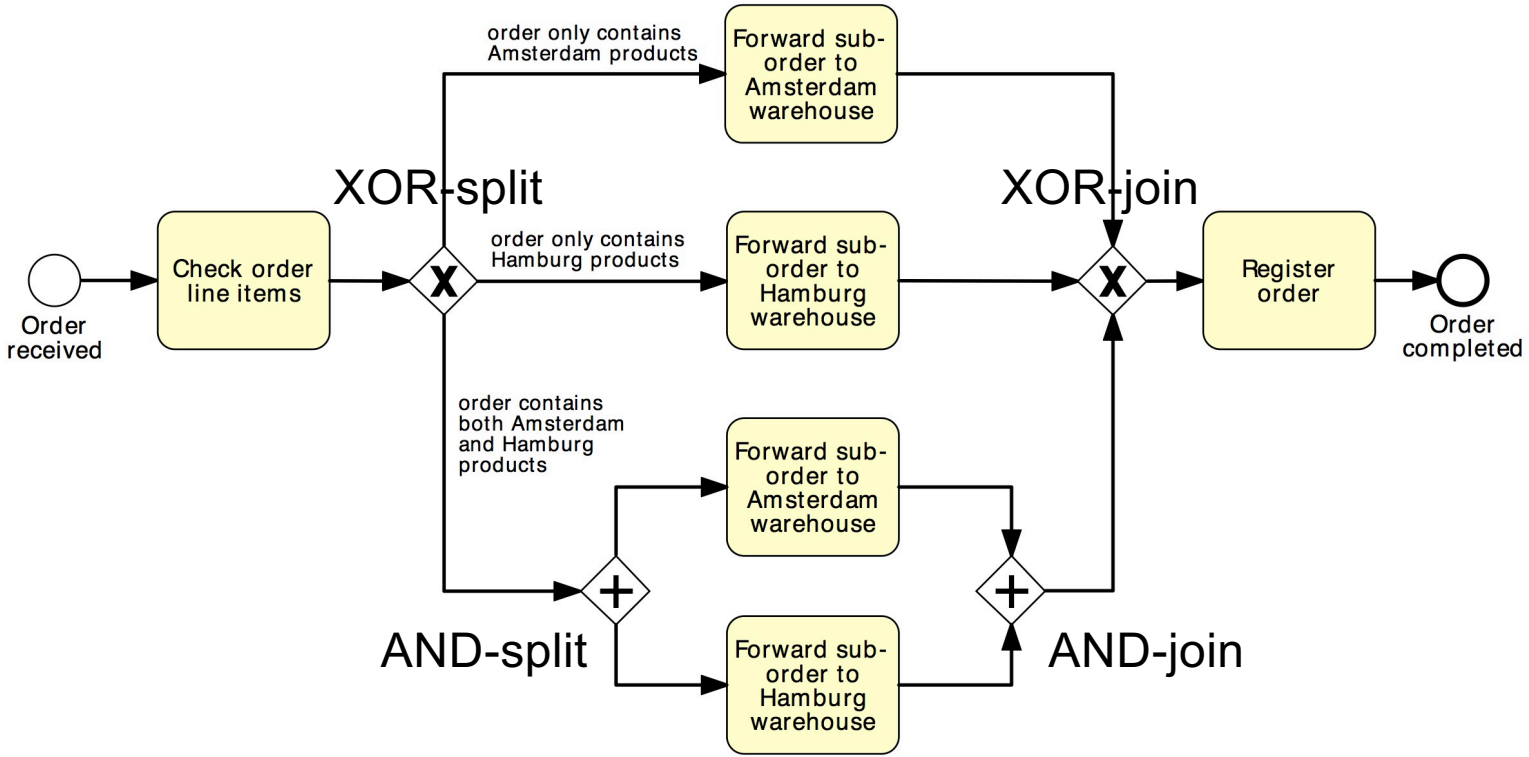
XOR / AND are not always what we need...

Order distribution process

A company has two warehouses, one in Amsterdam, the other in Hamburg, that store different products. When an order is received, it is distributed across these warehouses: if some of the relevant products are maintained in Amsterdam, a sub-order is sent there; likewise, if some relevant products are maintained in Hamburg, a sub-order is sent there. Afterwards, the order is registered and the process completes.

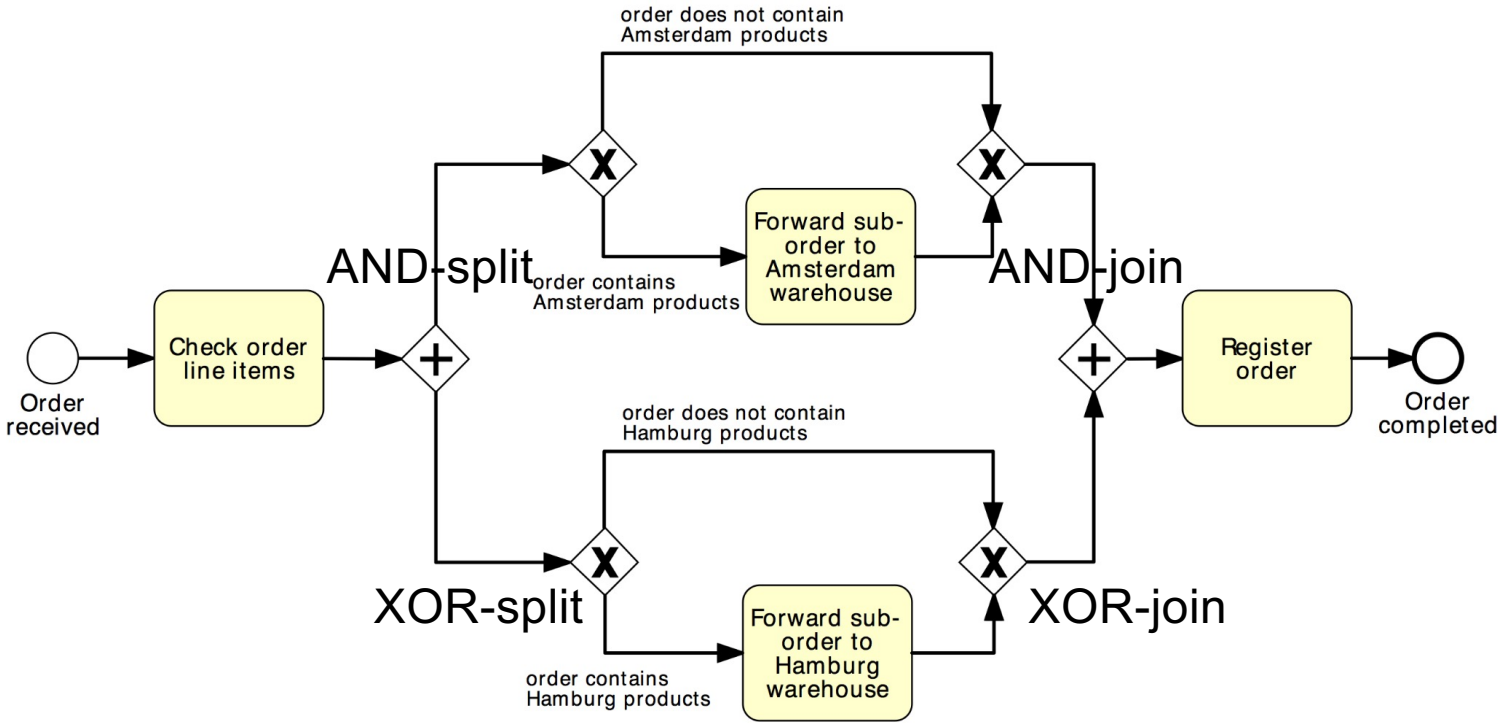
Solution 1

Order distribution process

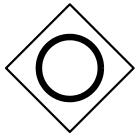


Solution 2

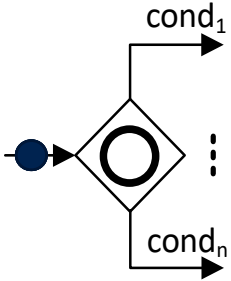
Order distribution process



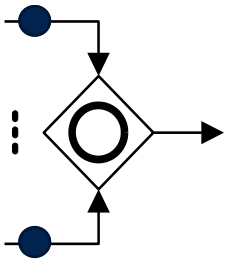
OR Gateway



An *OR Gateway* provides a mechanism to create and synchronize n out of m parallel flows.



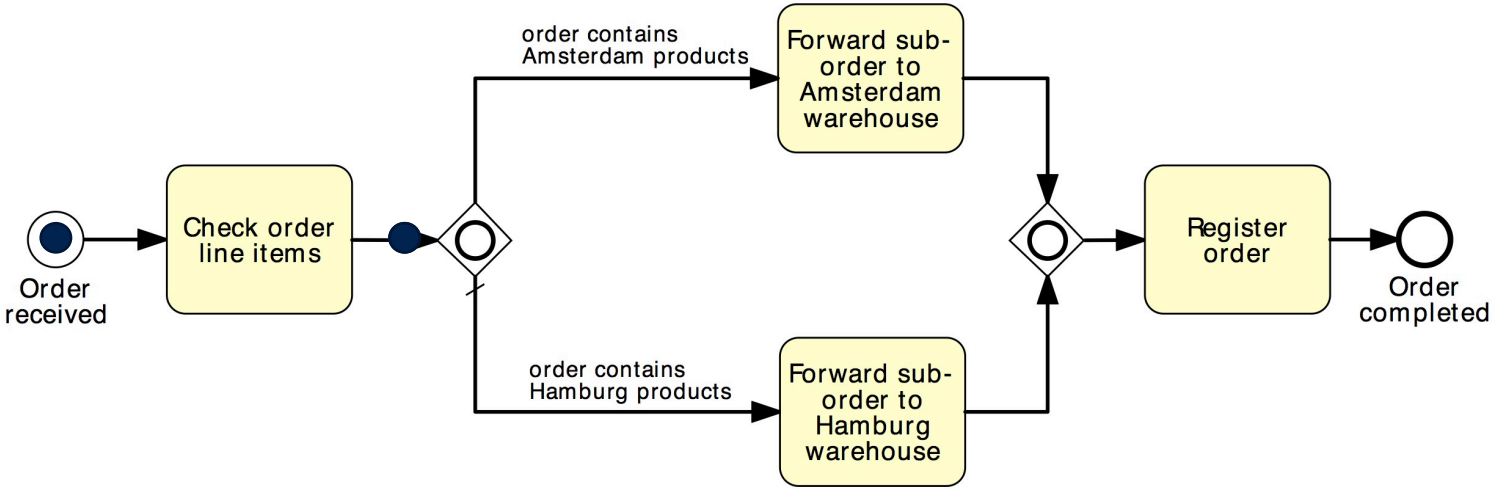
OR-split ➔ takes one or more branches depending on conditions



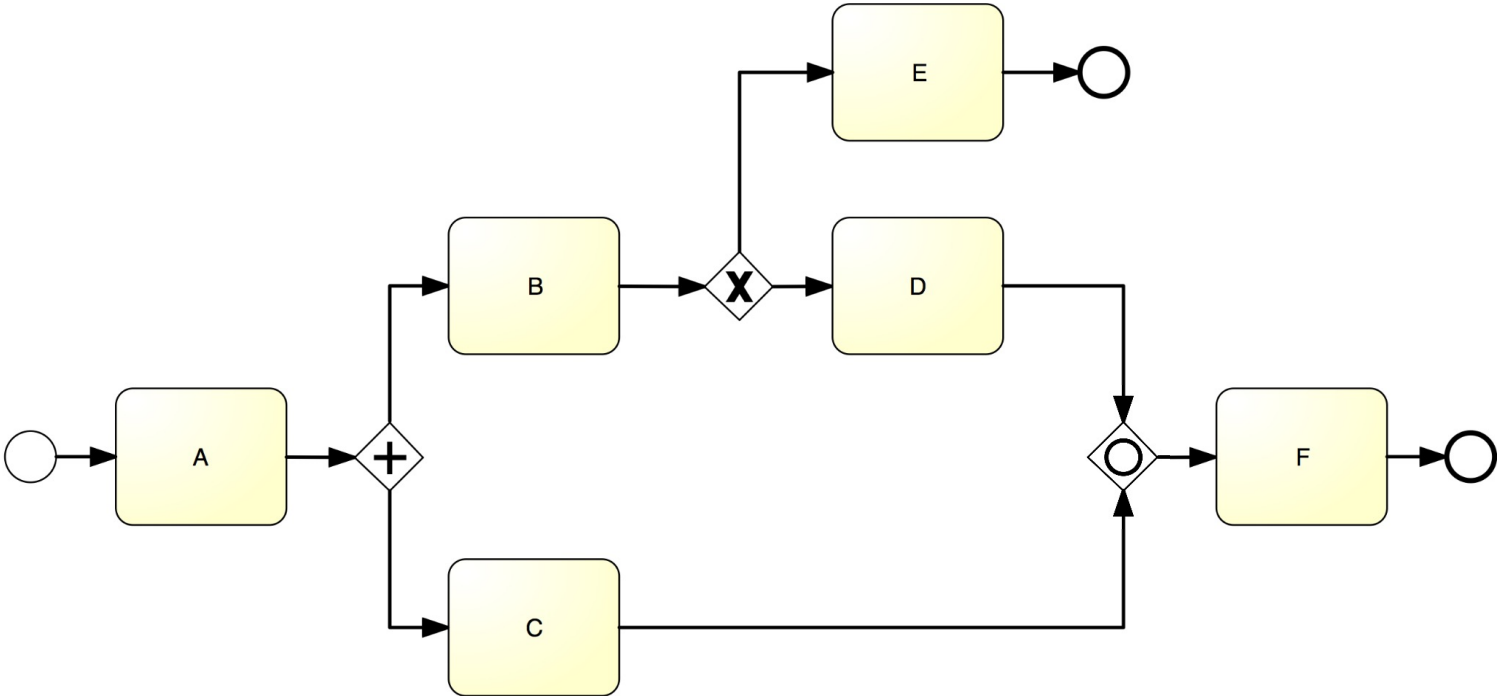
OR-join ➔ proceeds when all **active** incoming branches have completed

Solution using OR Gateway

Order distribution process



What join type do we need here?



Exercise 3.3

Model the following fragment of a business process for assessing loan applications.

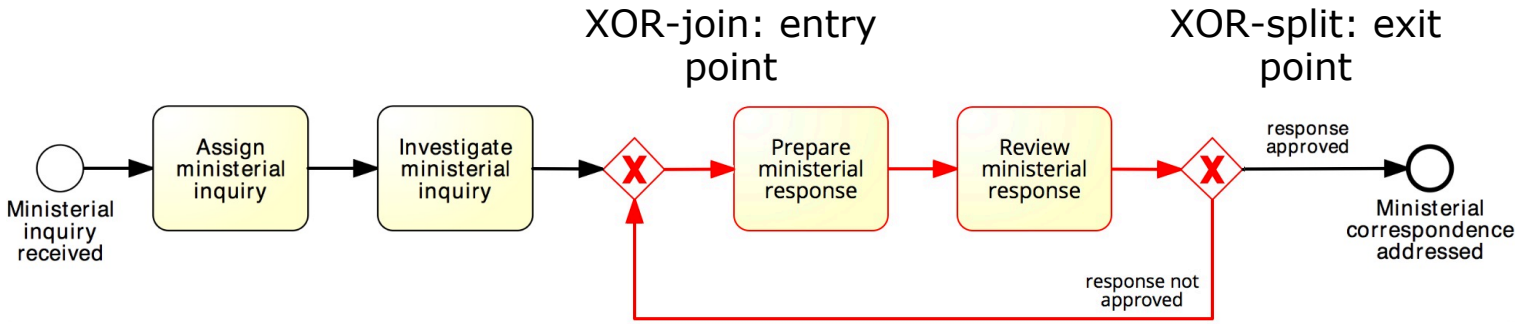
A loan application may be coupled with a home insurance which is offered at discounted prices. The applicants may express their interest in a home insurance plan at the time of submitting their loan application to the loan provider. Based on this information, if the loan application is approved, the loan provider may either only send an acceptance pack to the applicant, or also send a home insurance quote. The process then continues with the verification of the repayment agreement.

Rework and repetition

Address ministerial correspondence

In the treasury minister's office, once a ministerial inquiry has been received, it is registered into the system. Then the inquiry is investigated so that a ministerial response can be prepared.

The finalization of a response includes the preparation of the response itself by the cabinet officer and the review of the response by the principal registrar. If the registrar does not approve the response, the latter needs to be prepared again by the cabinet officer for review. The process finishes only once the response has been approved.

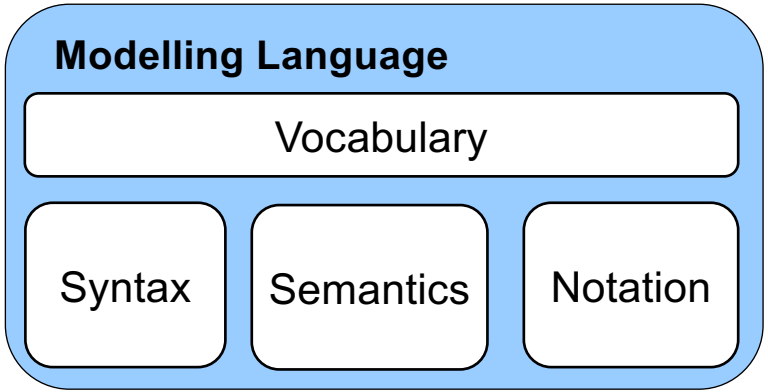


Exercise 3.4

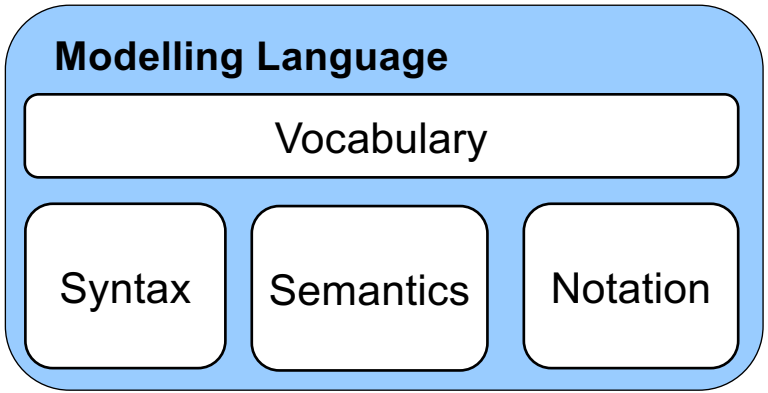
Model the following fragment of a business process for assessing loan applications.

Once a loan application is received by the loan provider, and before proceeding with its assessment, the application itself needs to be checked for completeness. If the application is incomplete, it is returned to the applicant, so that they can fill out the missing information and send it back to the loan provider. This process is repeated until the application is found complete.

Components of a modeling language

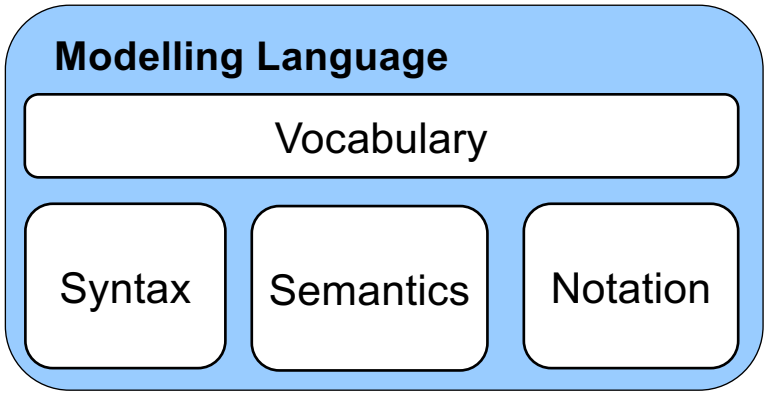


Components of a modeling language: Vocabulary



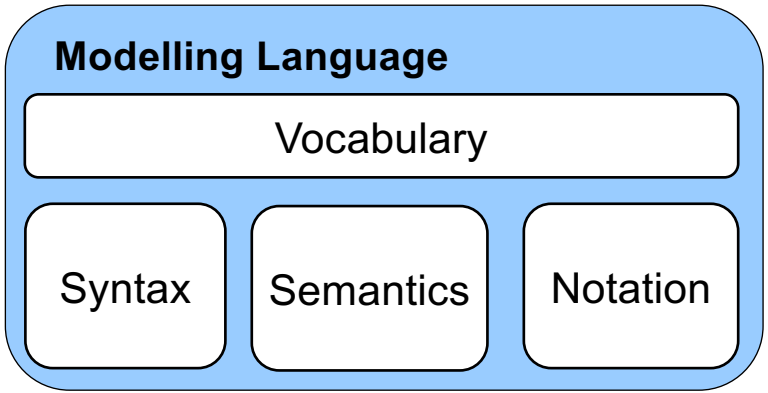
Vocabulary: set of modeling elements of the language (BPMN: activities, gateways, events…)

Components of a modeling language: Syntax



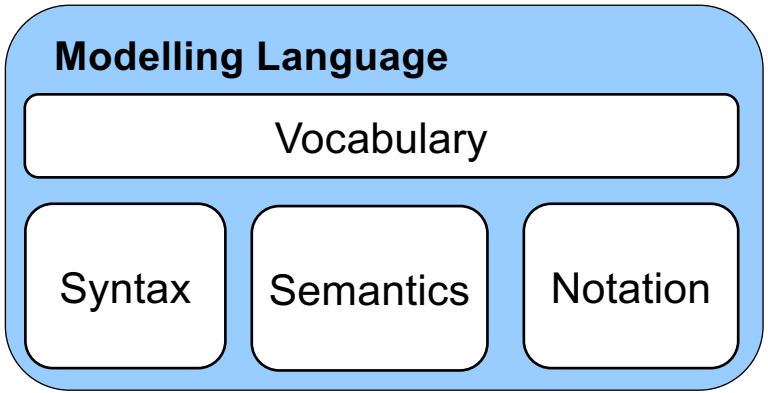
Syntax: set of rules to govern how these elements can be combined (BPMN: start events only have outgoing sequence flows whereas end events only have incoming sequence flows).

Components of a modeling language



Semantics: bind these elements, including their textual descriptions, to a precise meaning (in BPMN: activities model something actively performed during the business process, while XOR gateways model exclusive decisions and simple merging points).

Components of a modeling language: Notation



Notation: set of graphical symbols for the visualization of the elements (in BPMN: labeled rounded boxes to depict activities and the circles with a thin border to depict start events).

Business Objects (aka artifacts)

Can be:

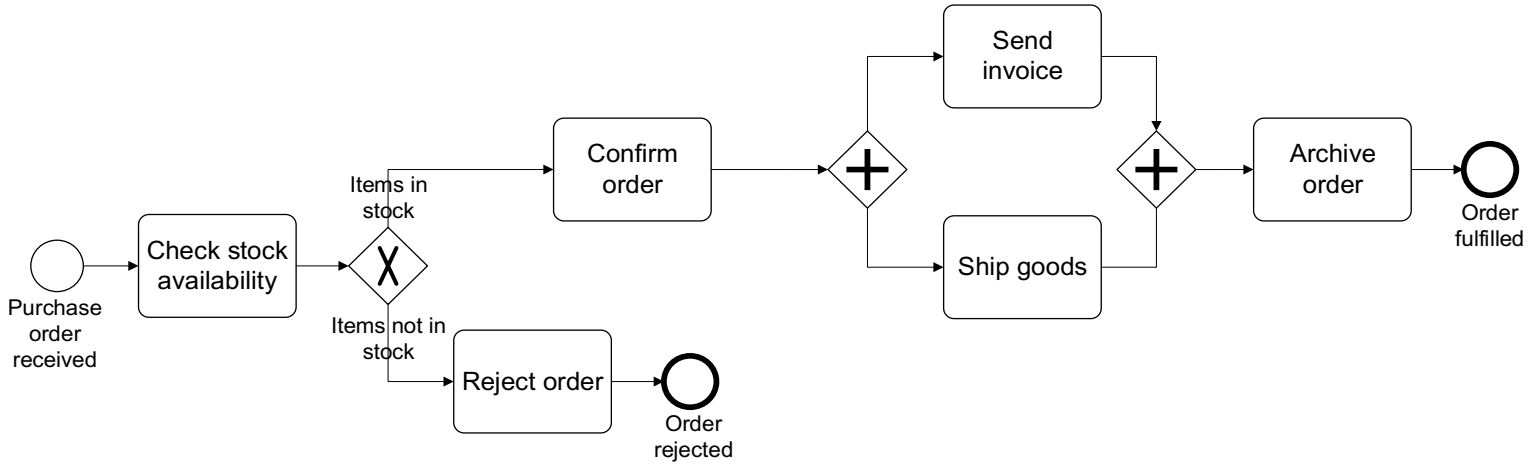
- Physical or digital information artifacts (e.g. an order on paper, an invoice on PDF)
- Physical material (e.g. a box containing the ordered goods)





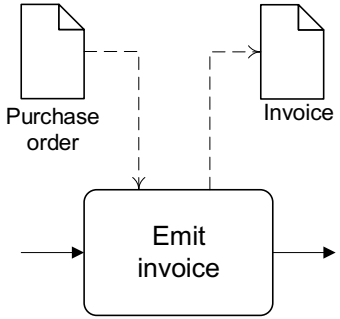


Our Order-to-cash process, again



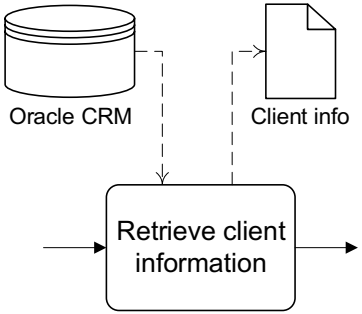
The purchase order document serves as an input to the stock availability check against the Warehouse DB. Based on the outcome of this check, the status of the document is updated, either to “approved” or “rejected”. If the order is approved, an invoice and a shipment notice are produced. The order is then archived on the Orders DB.

Business Objects in BPMN



A Data Object captures an artifact required (input) or produced (output) by an activity.

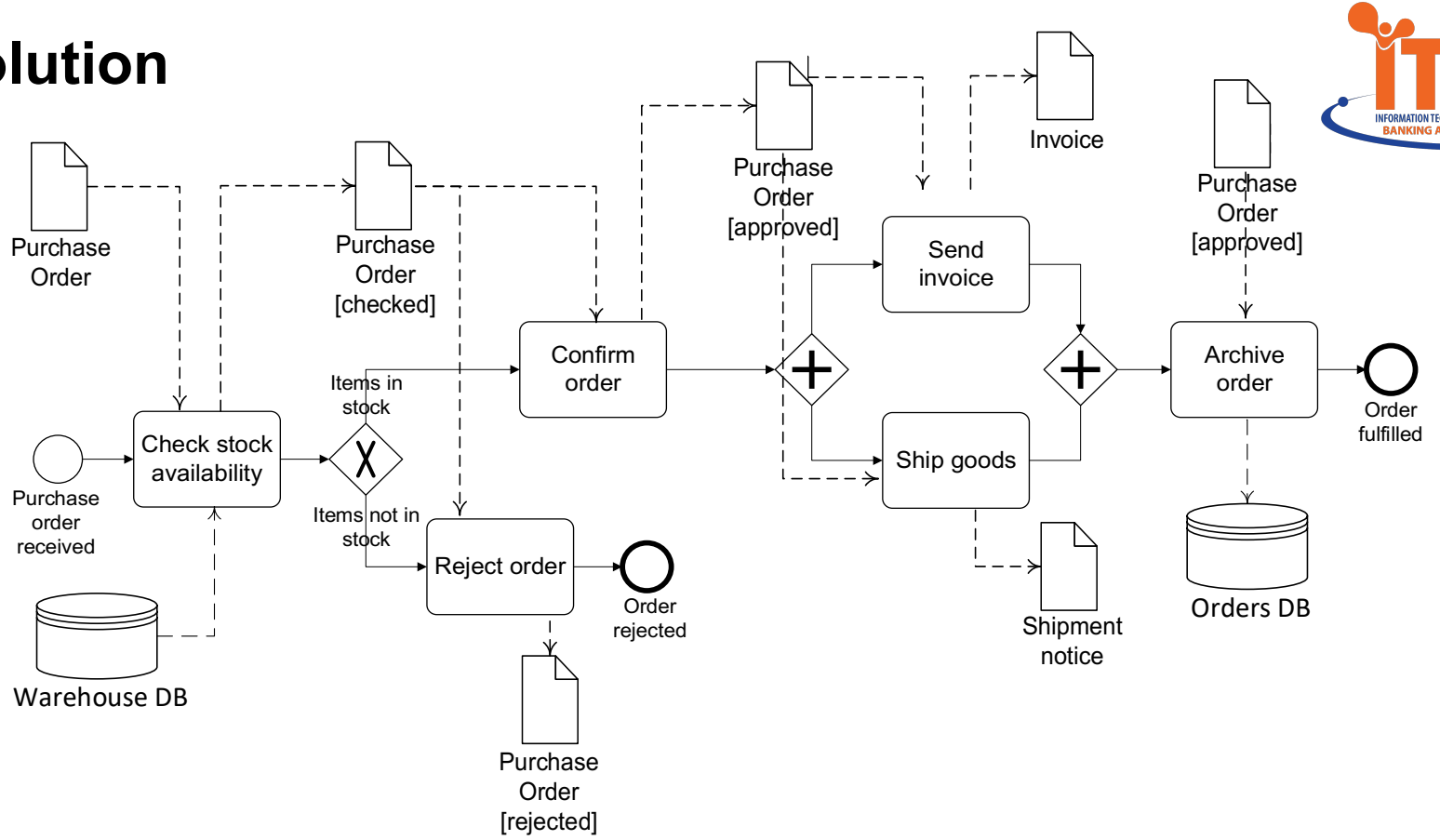
- Can be physical or electronic



A Data Store is a place containing data objects that must be persisted beyond the duration of a process instance.

It is used by an activity to store (as output) or retrieve (as input) data objects.

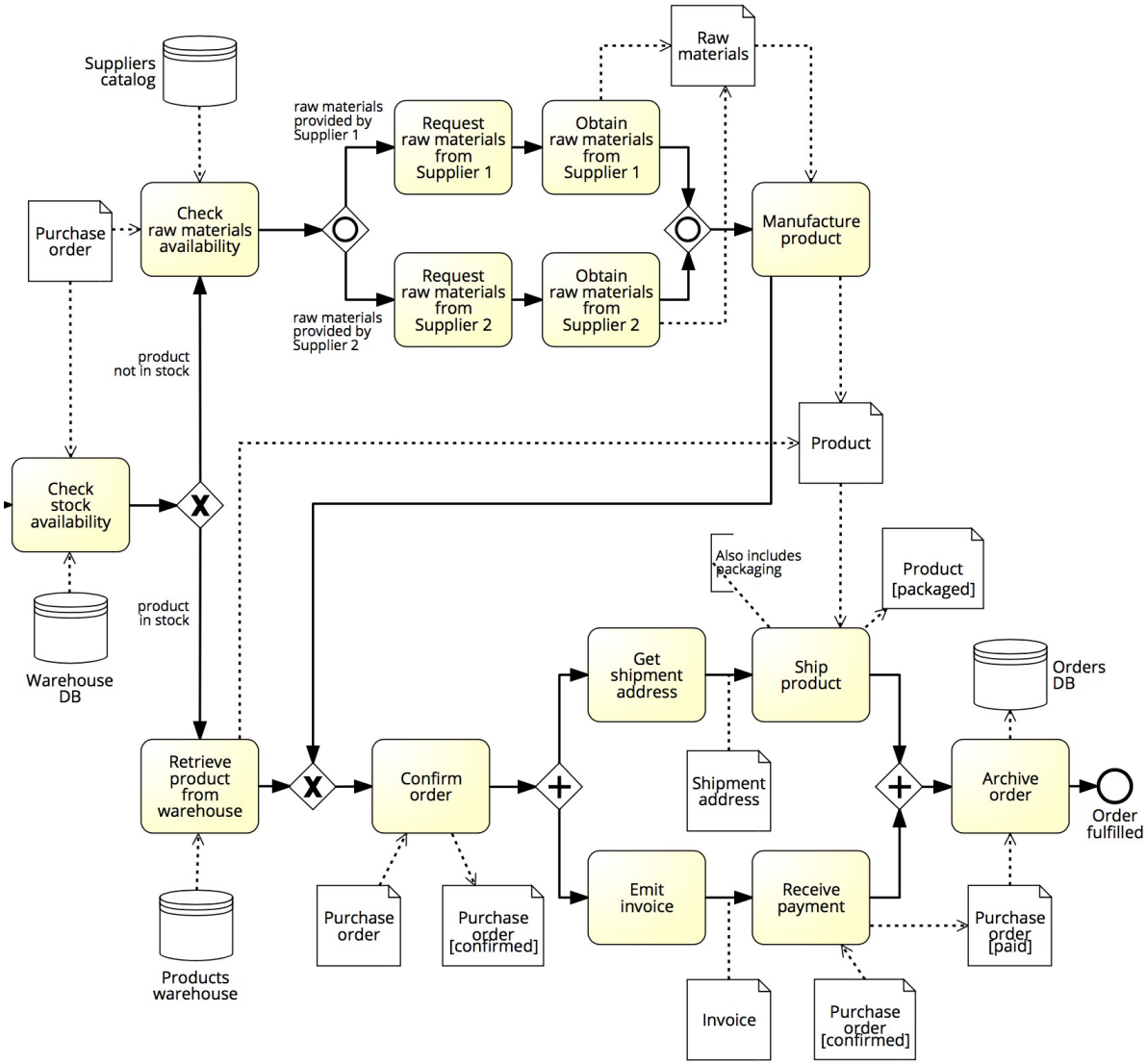
Solution



The purchase order document serves as an input to the stock availability check against the Warehouse DB. Based on the outcome of this check, the status of the document is updated, either to “approved” or “rejected”. If the order is approved, an invoice and a shipment notice are produced. The order is then archived on the Orders DB.

Exercise 3.5

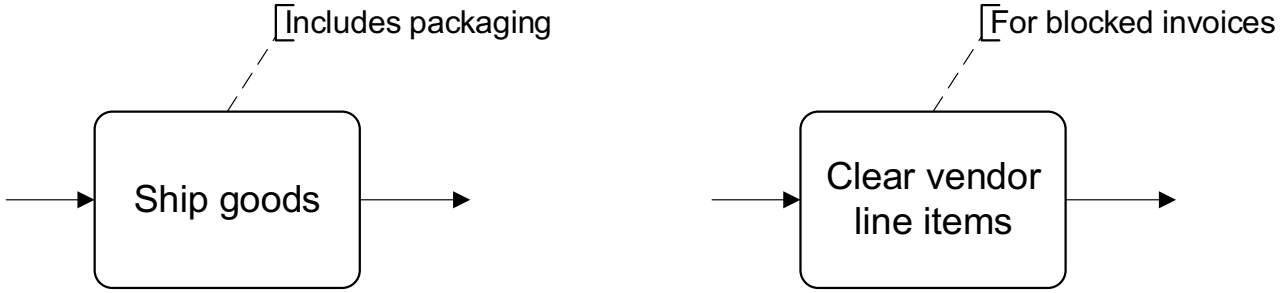
Is there any missing data object or data store in the example below?



BPMN Text Annotations

A *Text Annotation* is a mechanism to provide additional text information to the model reader

- **Doesn't affect** the flow of tokens through the process



Exercise 3.6

Put together the four fragments of the loan assessment process that you created in Exercises 3.1–3.4.

Hint. Look at the labels of the start and end events to understand the order dependencies among the various fragments. Then extend the resulting model by adding all the required business objects. Moreover, attach annotations to specify the business rules behind i) checking an application completeness, ii) assessing an application eligibility, and iii) verifying a repayment agreement.

Resources

Active resources:

- Process participant
- Software system
- Equipment

Resource class:

A **group of (active) resources** that are interchangeable, e.g. a role, an organizational unit or the whole organization.



Resources in the order-to-cash example

The order-to-cash process is carried out by a seller's organization which includes two departments: the **Sales department** and the **Warehouse & Distribution department**.

The purchase order received by the Seller has to be checked against the stock. This is done via an **ERP** module within the **Warehouse & Distribution department**.

If the purchase order is confirmed, the **Warehouse & Distribution department** ships the goods. Meantime, the **Sales department** emits the invoice. The process concludes with the order being archived by the **Sales department**.

BPMN Elements – Pools & Lanes

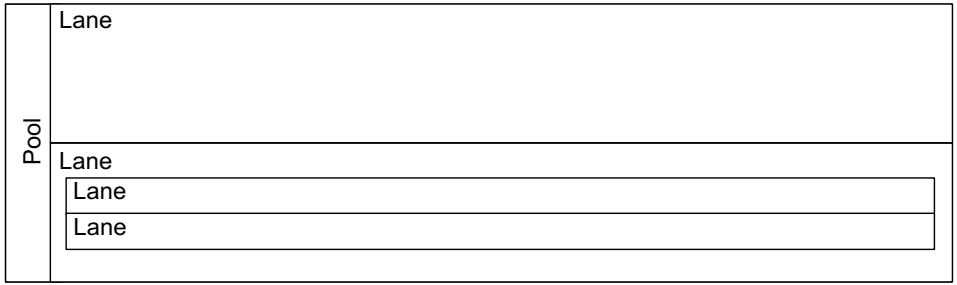
Pool

Captures a resource class. Generally used to model a business party (e.g. a whole company)

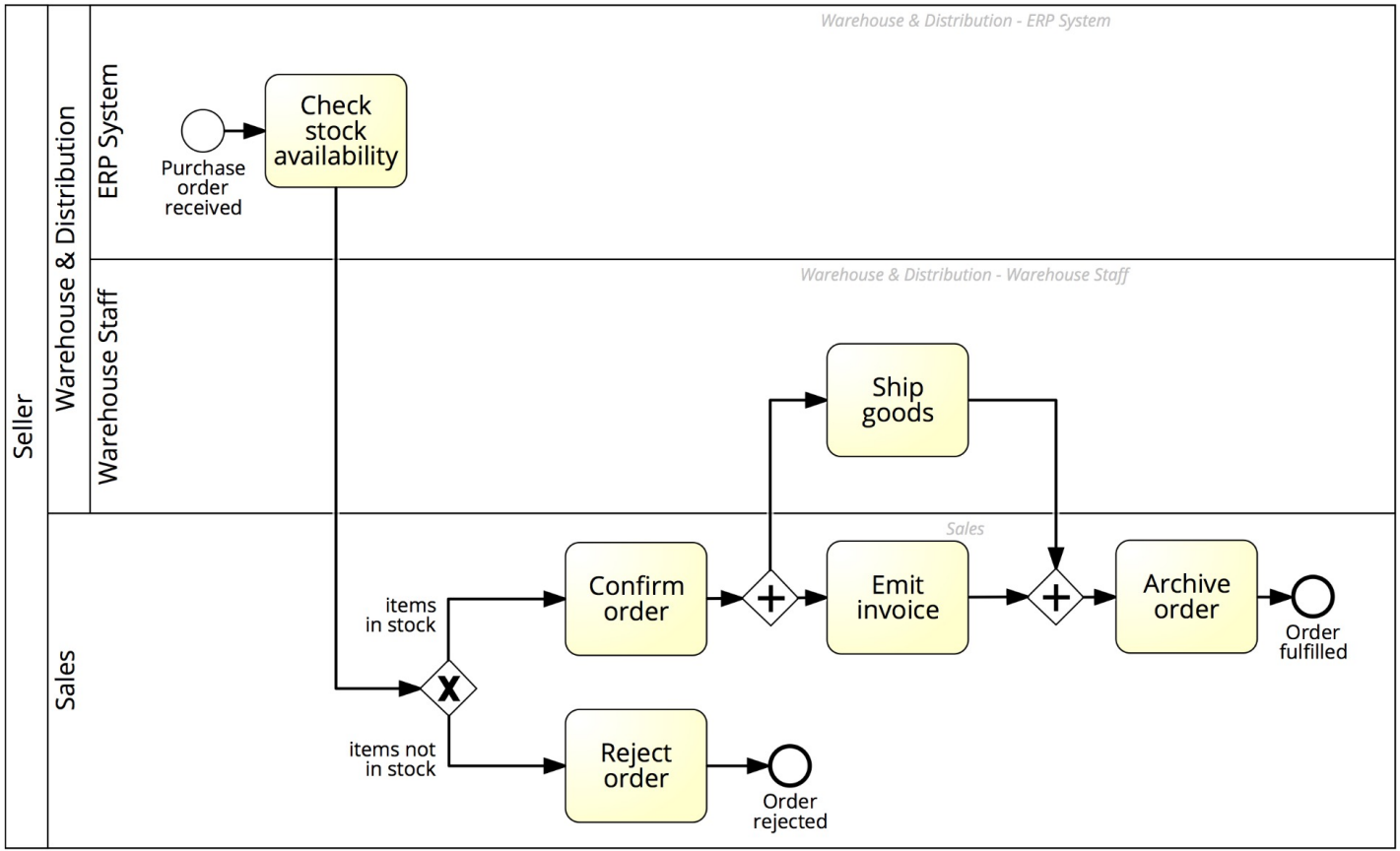


Lane

Captures a *resource sub-class* within a resource class by partitioning a pool. Generally used to model departments (e.g. shipping, finance), internal roles (e.g. Manager, Associate), software systems (e.g. DBMS, CRM) or equipment (e.g. Manufacturing plant)



Solution: Order-to-cash



Exercise 3.7

Extend the business process for assessing loan applications that you created in Exercise 3.6 by considering the following resource aspects.

The process for assessing loan applications is executed by four roles within the loan provider: a financial officer takes care of checking the applicant's credit history; a property appraiser is responsible for appraising the property; an insurance sales representative sends the home insurance quote to the applicant if this is required. All other activities are performed by the loan officer who is the main point of contact with the applicant.

Exchanging information between business parties

Order-to-cash

The purchase order **sent by the Customer** is received by the Seller and checked against the stock. This is done via an ERP module within the Warehouse & Distribution department. If the purchase order is not confirmed, the Sales department **sends an order rejection to the Customer**, otherwise it **sends an order confirmation**.

Next, the Warehouse & Distribution department ships the goods and **sends a shipment notification to the Customer**. Meantime, the Sales department emits the invoice and **sends it to the Customer**. The process concludes with the order being archived by the Sales department.

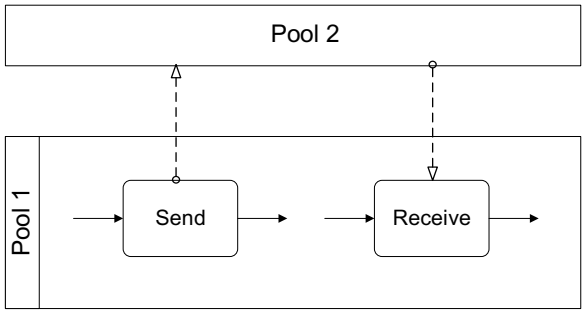
BPMN Elements – Message Flow

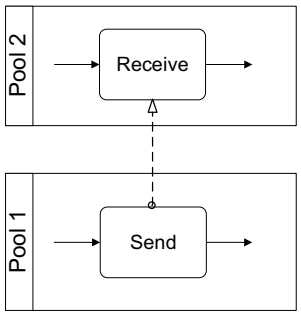
A *Message Flow* represents a flow of information or materials between two process parties (Pools)



A Message Flow can connect:

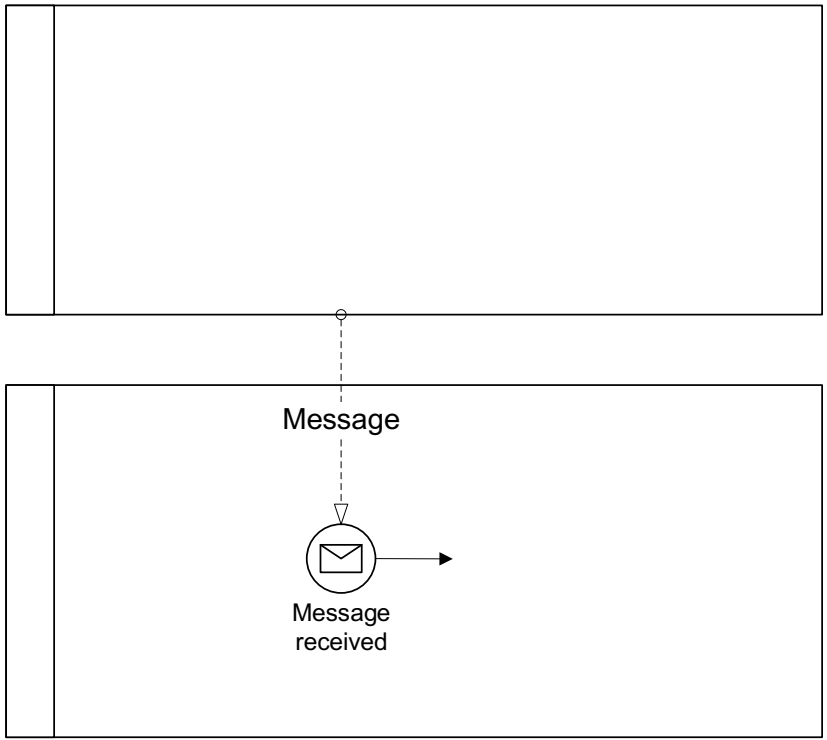
- directly to the boundary of a Pool ➔ captures a message to/from that party
- to a specific activity or event within that Pool ➔ captures a message that triggers a *specific* activity/event within that party



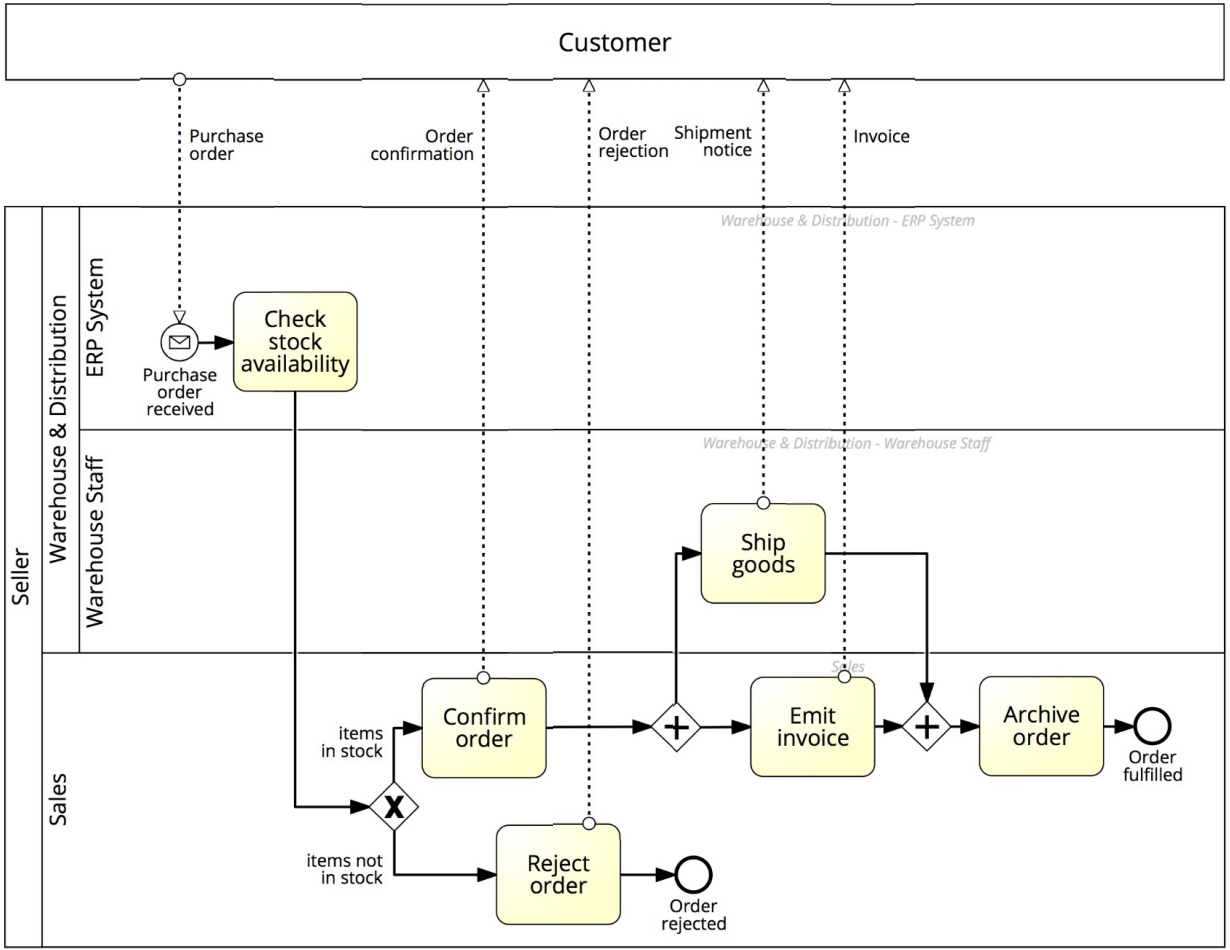


BPMN Elements – Start Message Event

The *start message event* triggers a process by the receipt of a message when an incoming message flow is connected to the event

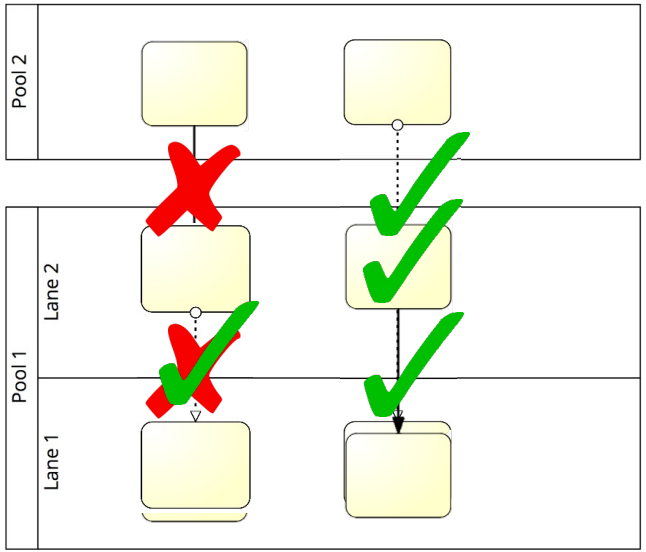


Solution: Order-to-cash



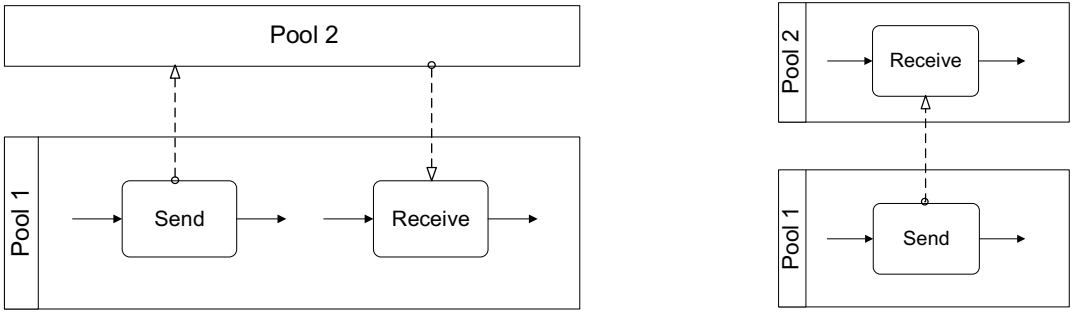
Pools, Lanes and Message Flows: syntax

1. The Sequence Flow **cannot** cross the boundaries of a Pool
2. Both Sequence Flow and Message Flow **can cross** the boundaries of Lanes
3. A Message Flow **cannot connect** two flow elements within the same pool



When are messages sent or received?

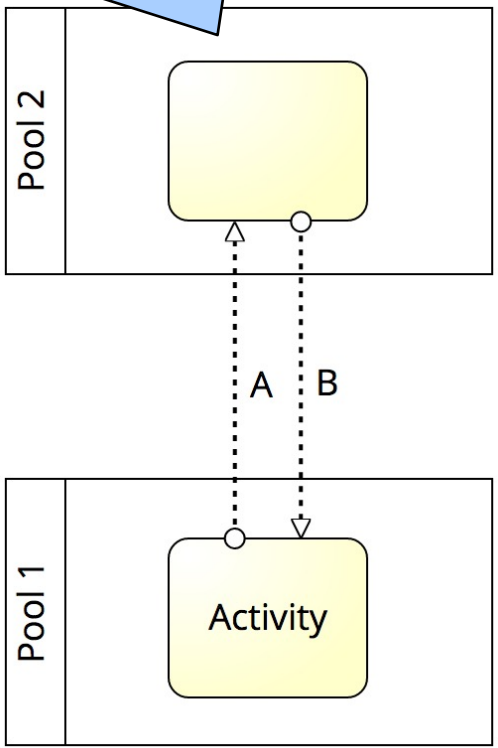
- A **Send** activity will send the outgoing message upon activity completion
- A **Receive** activity won't start until the incoming message has been received



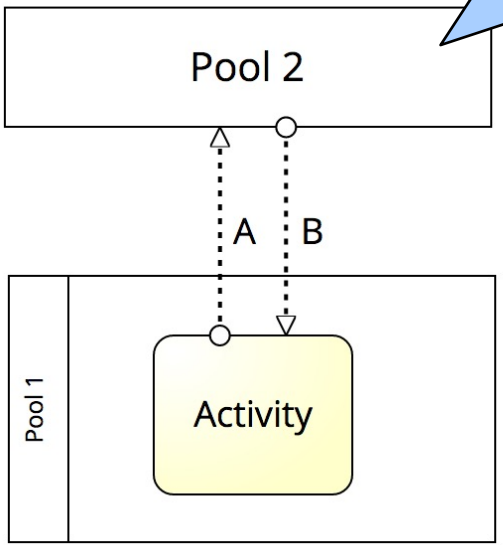
Note: the order of the message flows w.r.t. an activity is irrelevant, the above rules always hold

When are messages sent or received?

- Message B is first received before Activity can start.
- Message A is sent after, upon Activity's completion



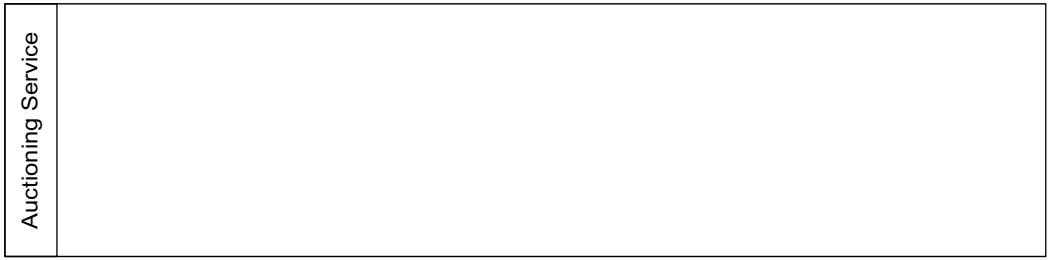
- First, message B is received, before Activity can start.
- Then, message A is sent, upon Activity's completion



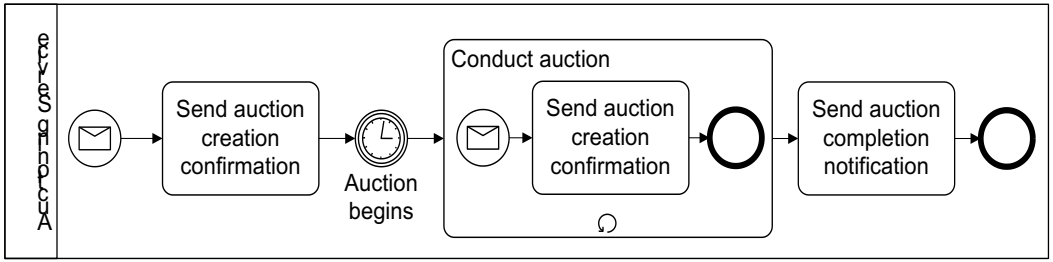
Process (or Orchestration) Diagram

Models a single business party and can be:

Public view (*black box*)

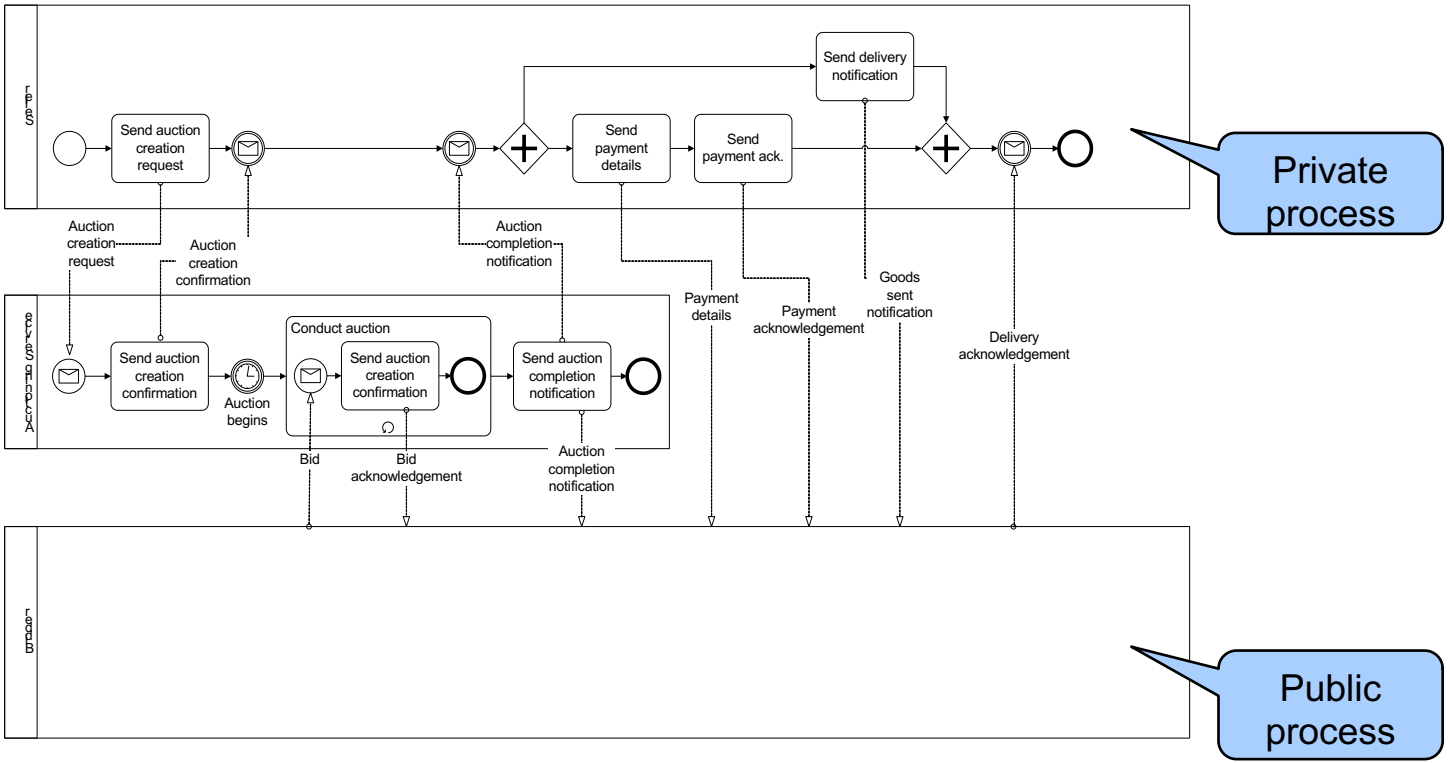


Private view (*white box*)



Collaboration Diagram

Models a global business process between at least two business parties (each modelled by a Pool)

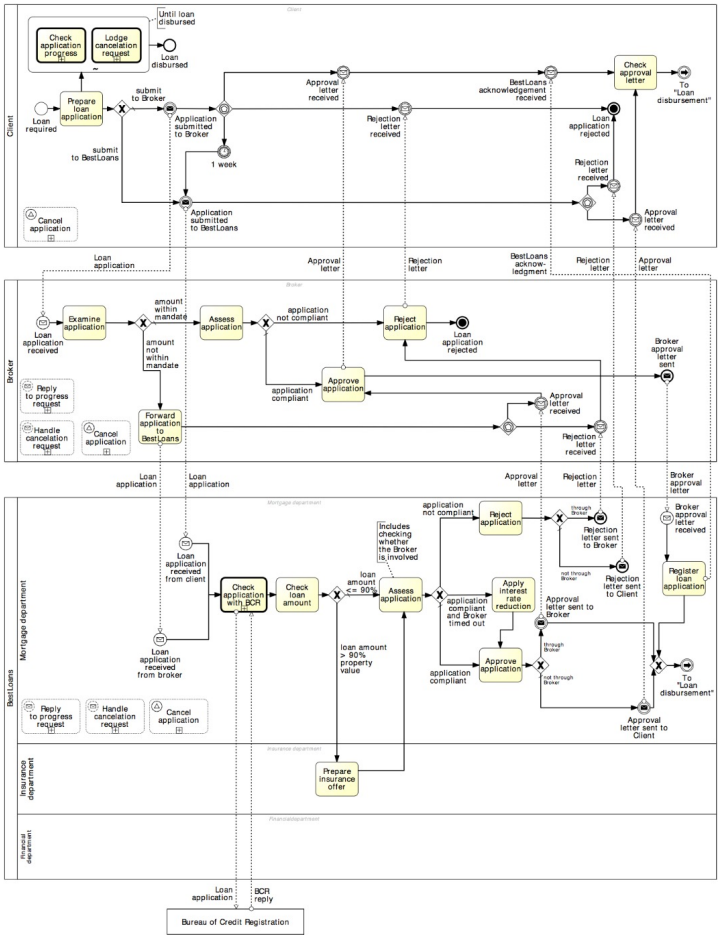


Exercise 3.8

Extend the model of Exercise 3.7 by representing the interactions between the loan provider and the applicant.

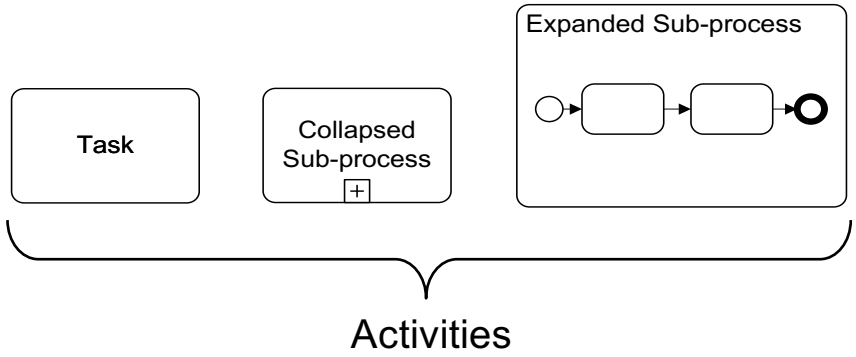


Process decomposition



Process decomposition

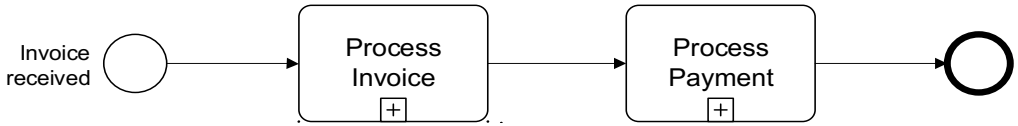
An activity in a process can be decomposed into a “sub-process”



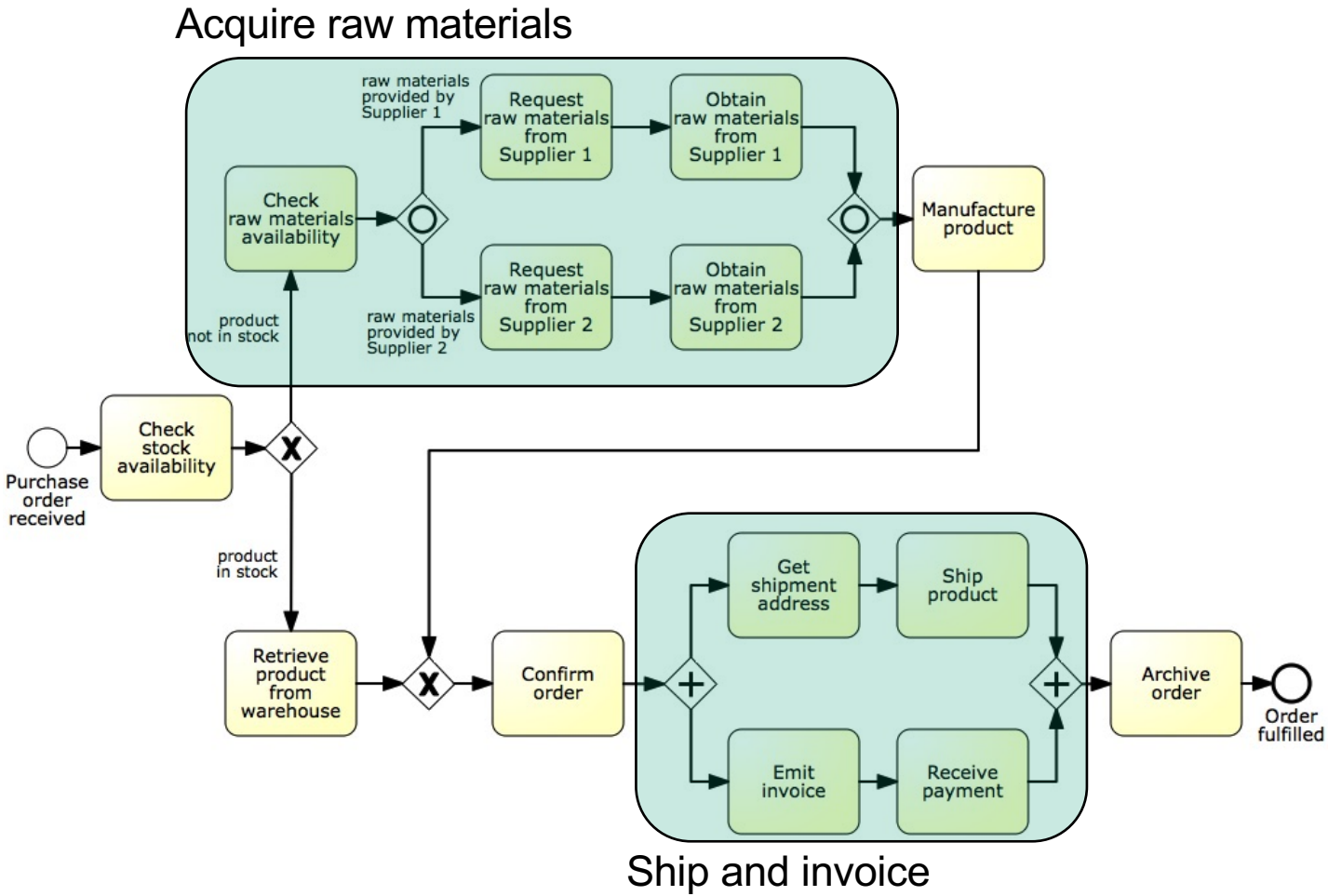
Use this feature to:

1. Improve understanding by breaking down large models
2. Identify parts that should be:
   - repeated
   - executed multiple times in parallel
   - interrupted, or
   - compensated

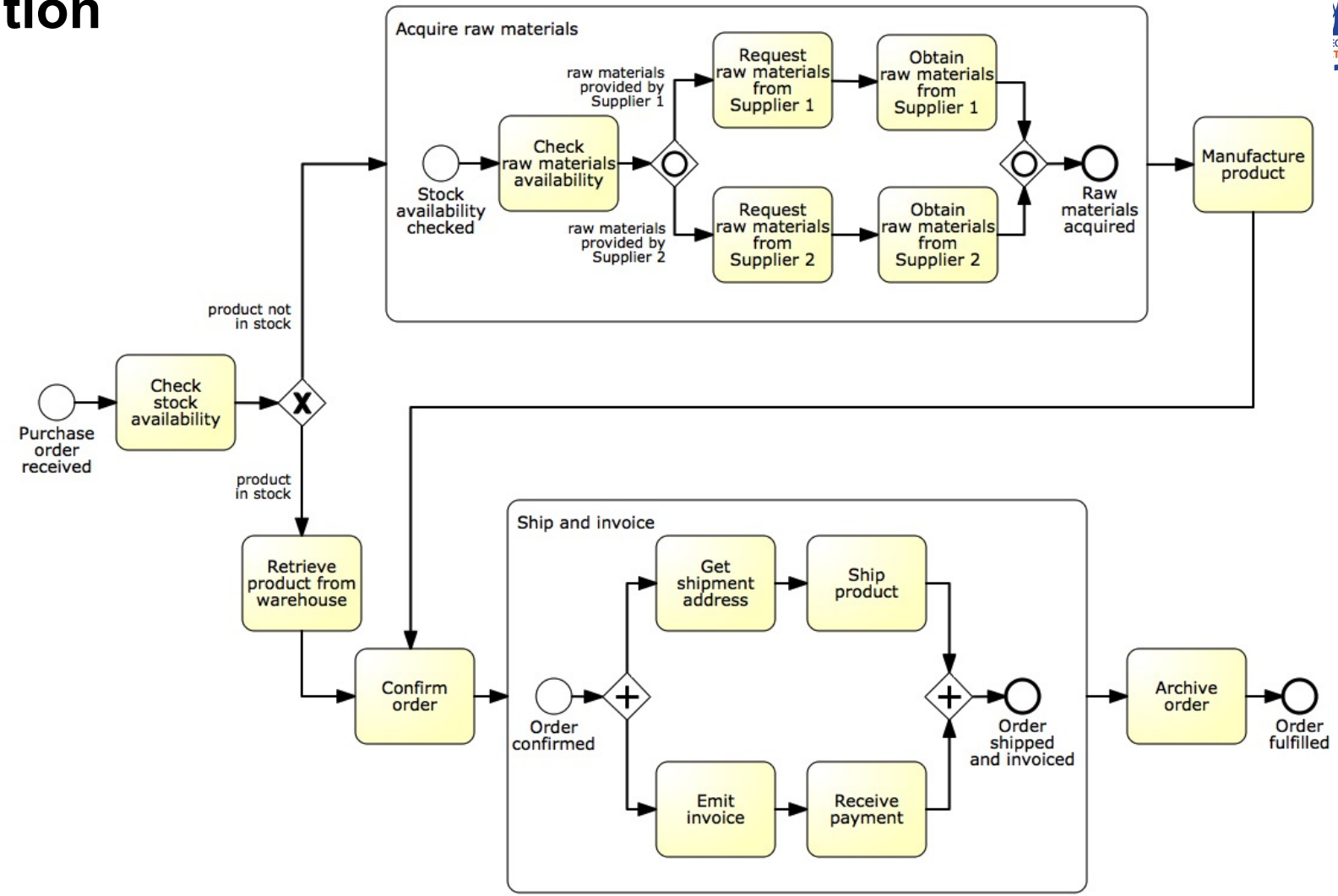
Example: Sub-Process



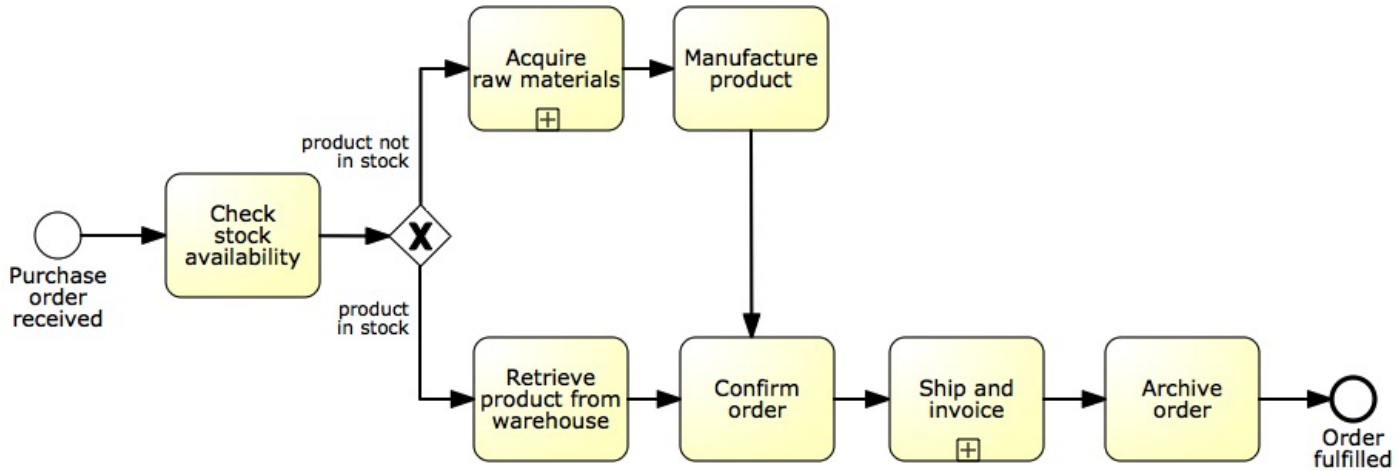
Identify possible sub-processes



Solution


Acquire raw materials
raw materials provided by Supplier 1
Request raw materials from Supplier 1
Obtain raw materials from Supplier 1
Check raw materials availability
Stock availability checked
raw materials provided by Supplier 2
Request raw materials from Supplier 2
Obtain raw materials from Supplier 2
Raw materials acquired
Manufacture product
product not in stock
Check stock availability
Purchase order received
product in stock
Retrieve product from warehouse
Confirm order
Ship and invoice
Get shipment address
Ship product
Order confirmed
Order shipped and invoiced
Emit invoice
Receive payment
Archive order
Order fulfilled

The refactored model



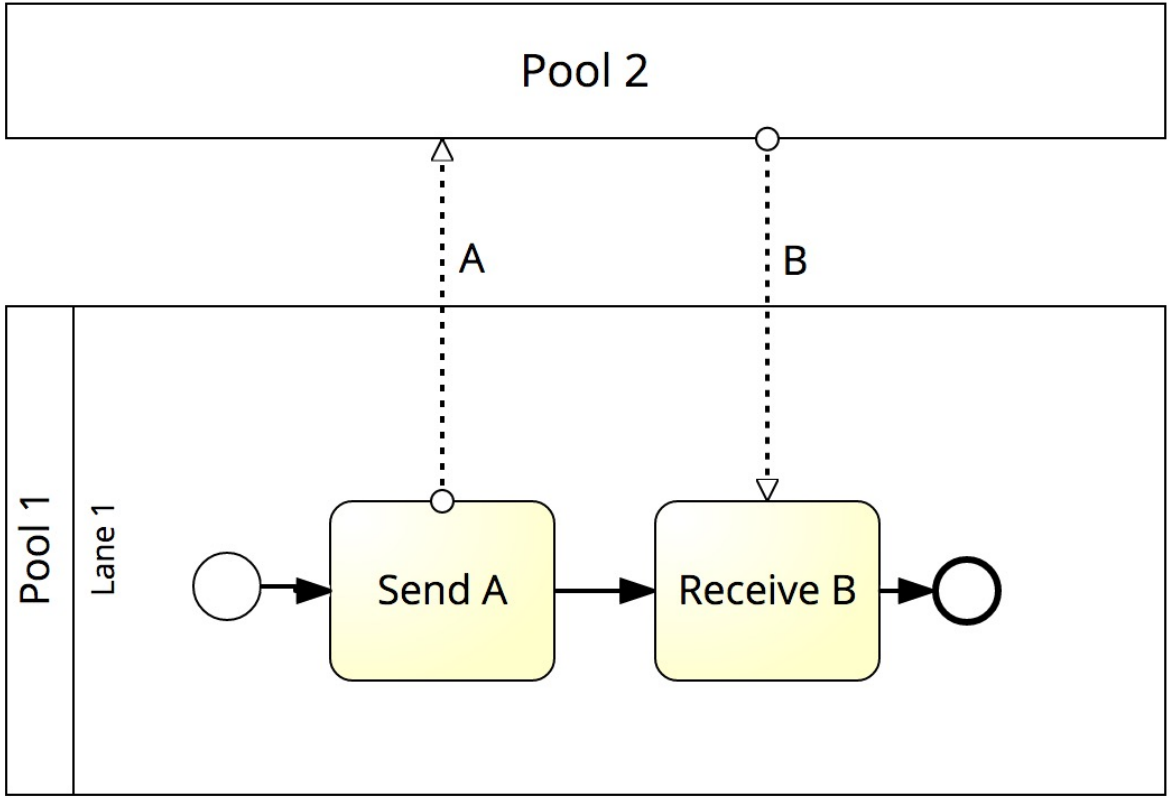
Exercise 3.9

Identify suitable sub-processes in the process for assessing loan applications modeled in Exercise 3.6.

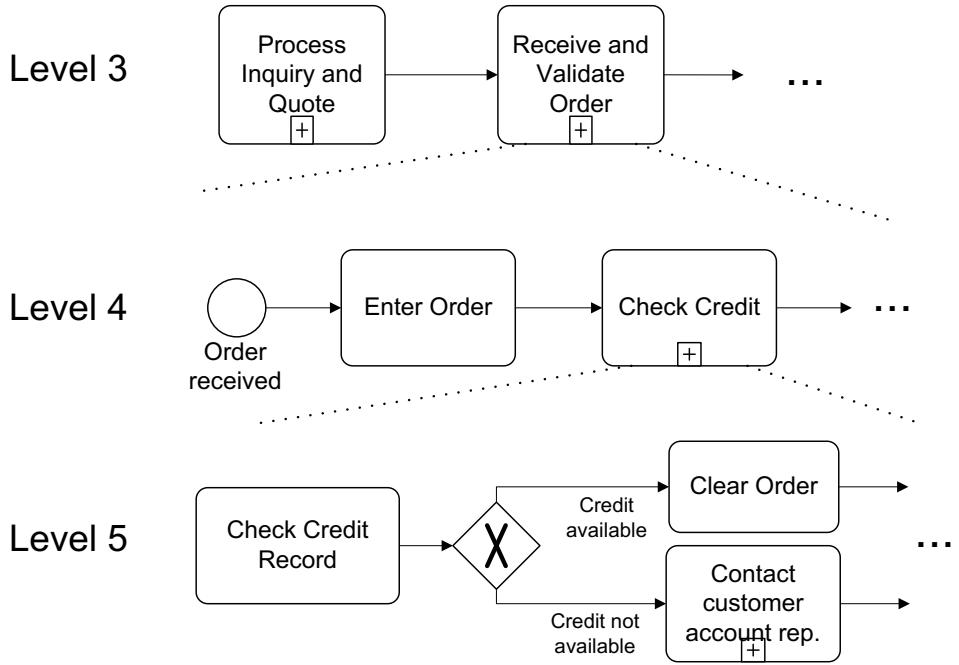
Hint. Use the building blocks that you created throughout Exercises 3.1–3.4.

Imposing order of messages via subprocess

The expanded subprocess for “Activity”



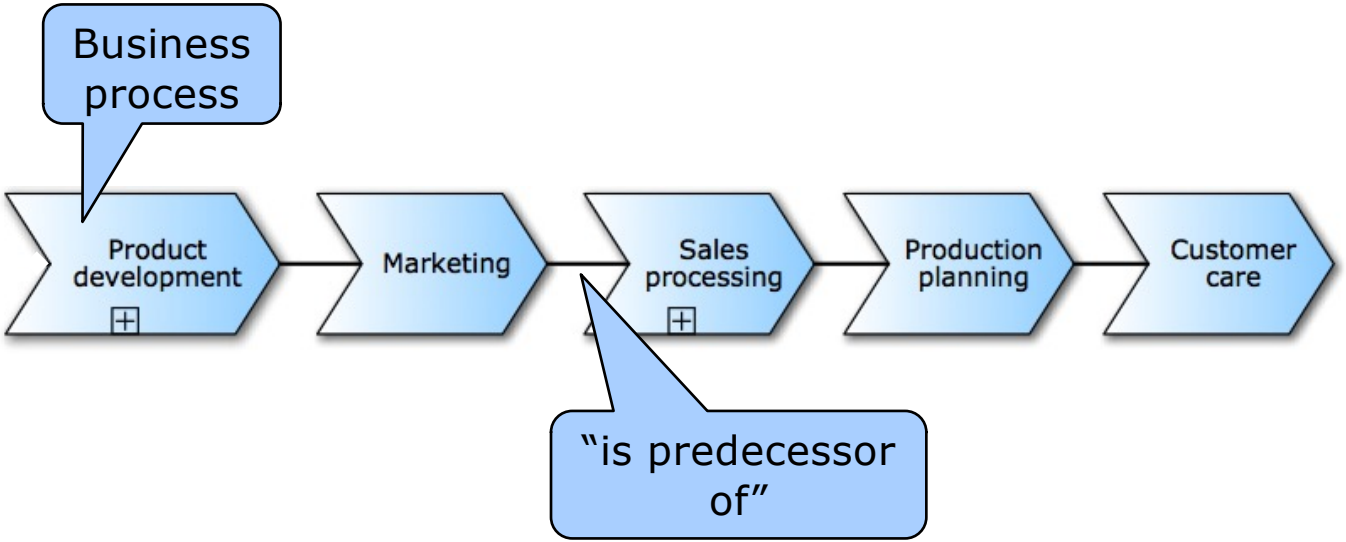
Example: Modelling process hierarchies



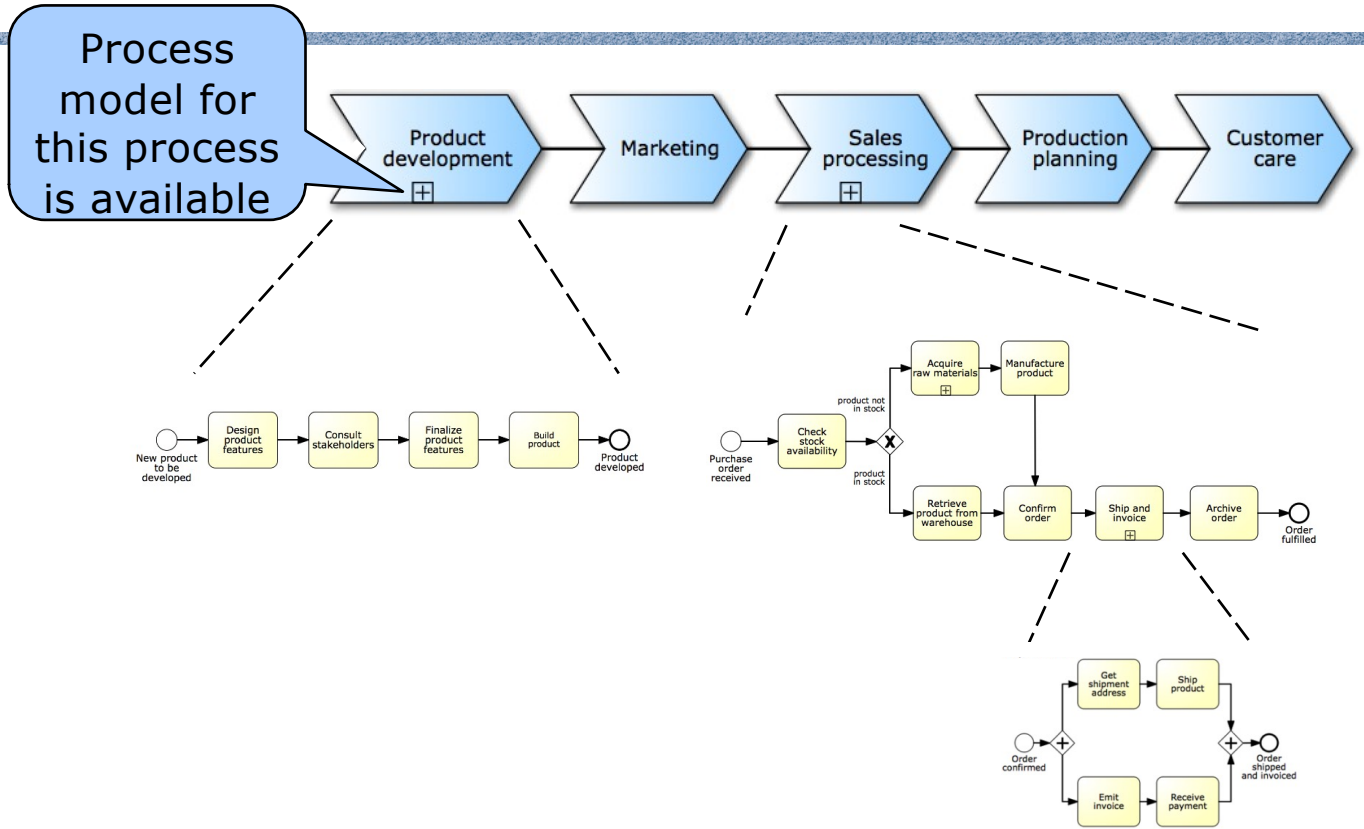
(Fragment of the SCOR reference model)

Value chain modelling

Chain of (high-level) processes an organization performs in order to achieve a business goal, e.g. deliver a product or service to the market.



Linking value chains with process models



Guidelines: modeling levels

Use sub-processes when the model becomes too large:

- Hard to understand
- Increased error probability

Level 1 – start with value chain

Level 2 – add main decisions and handoffs (lanes)

Level 3+ – add procedural aspects:

- Parallel gateways
- Data objects, data stores
- Exception handling
- And as much detail as is **relevant**

Decomposition drivers:

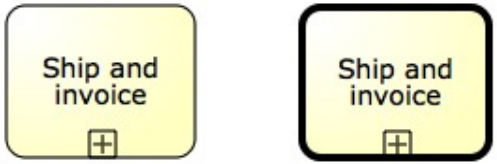
- Logical: group elements meaningfully (e.g. common business object)
- Structural: up to **30** nodes (activities, events, gateways)

Process Reuse

By default, a sub-process is “embedded” into its parent process (i.e. it is stored within the same file)

In order to maximize reuse, it is possible to “extract” the sub-process and store it as a separate file in the process model repository

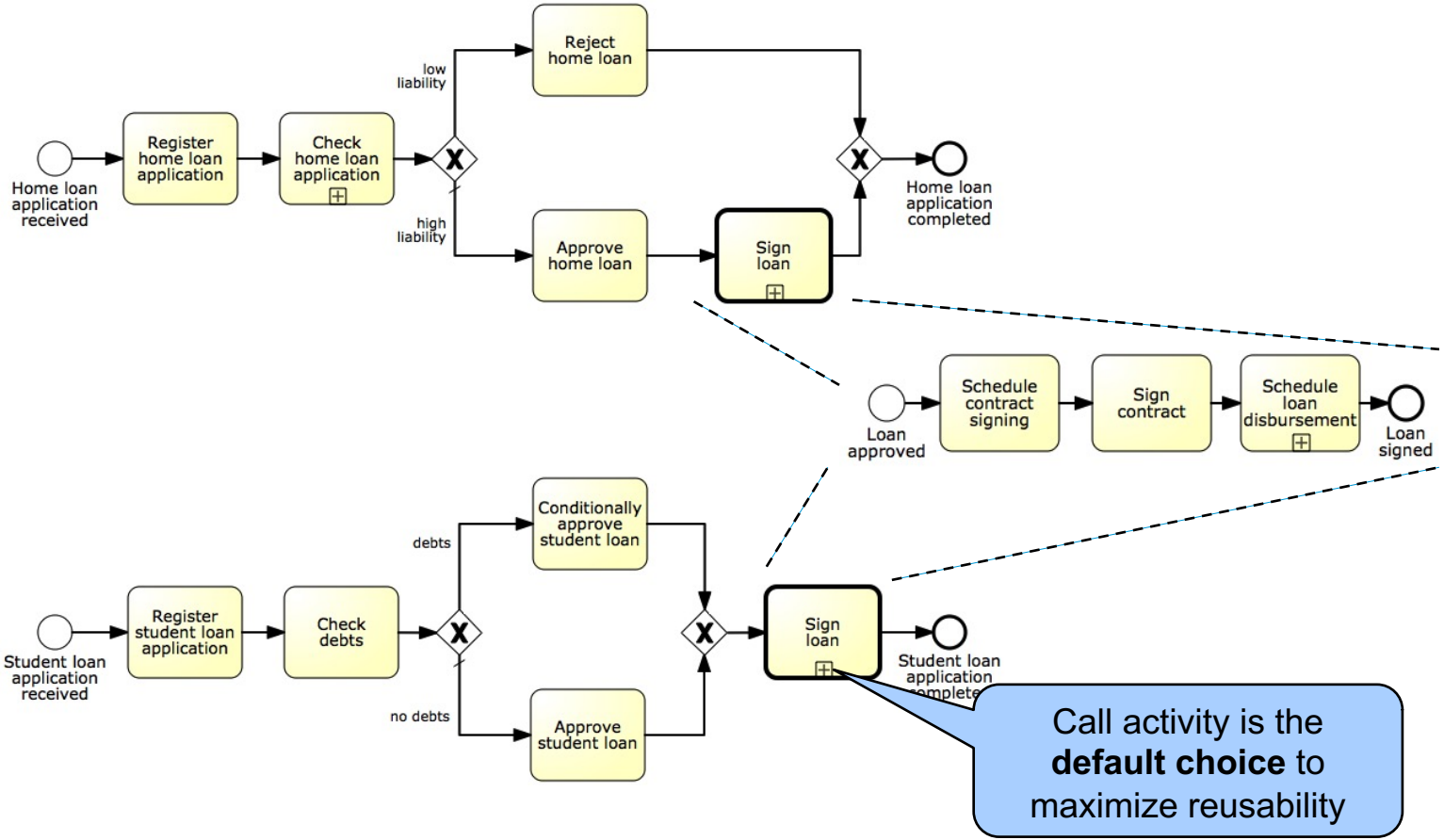
Such a sub-process is called “global” model, and is invoked via a “call” activity



(normal) activity

Call activity

Example: process reuse



Exercise 3.9

Identify suitable sub-processes in the process for assessing loan applications modeled in Exercise 3.6.

Hint. Use the building blocks that you created throughout Exercises 3.1–3.4.

Sub-processes: syntax

Sequence flows cannot cross sub-process boundaries

- Use start/end events

Message flows can cross sub-process boundaries

- To indicate messages emanating from/incoming into the sub-process

Start with at least one start event

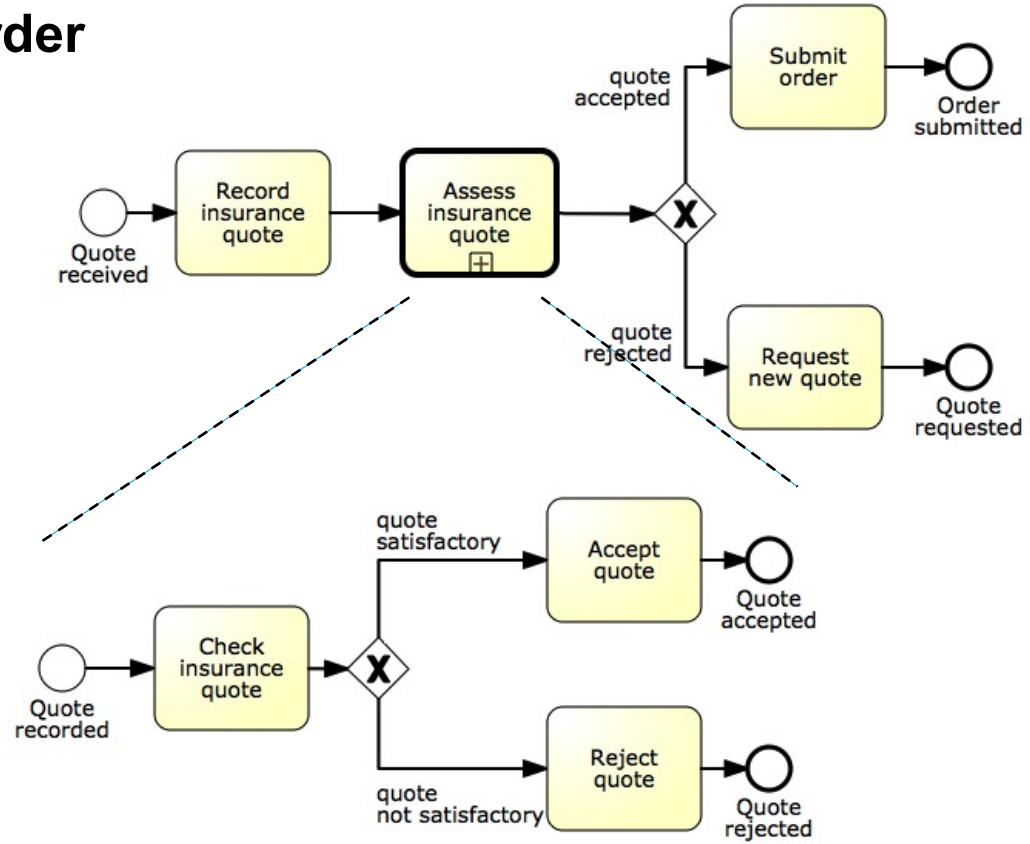
- If multiple, first occurring will trigger the sub-process

Finish with at least one end event

- The sub-process will complete once all tokens have reached an end event. May need an (X)OR-split after sub-process to understand what end event(s) have been reached

Example: sub-process with multiple end events

Quote-to-order



Exercise 3.10

Identify suitable sub-processes in the process of Exercise 1.7 (page 30). Among these sub-processes, identify those that are specific to this process versus those that can potentially be shared with other processes of the same company.

Recap

1. BPMN activities capture units of work in a process
2. Events define the start and end of a process, and signal something that happens during the execution of it
3. Gateways model exclusive and inclusive decisions, merges, parallelism and synchronization, and repetition
4. A process model depicts all the possible ways a given business process can be executed, while a process instance captures one specific process execution out of all possible ones
5. Data objects capture a physical or digital business object required to execute an activity or trigger an event, or that results from the execution of an activity or an event occurrence
6. Pools generally model resource classes while lanes are used to partition pools
7. Sub-processes represent activities that can be broken down in a number of internal steps, as compared to tasks, which capture single units of work